CE2005
Linear Algebra

Coordinates, Rank, and Linear Transformations

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Intended Learning Outcomes

By the end of this lecture, you should be able to:

- Determine coordinate vectors and construct transition matrices between ordered bases.
- Find bases for matrix subspaces and use rank and nullity to analyze linear systems.
- Verify linearity and determine the kernel, image, and range of a transformation.
- Construct matrix representations with respect to specified bases and apply geometric transformations.
- Relate matrix representations of the same operator through similarity and a change of basis.

Outline

1. How Do Coordinates Change with an Ordered Basis?
2. How Do Row and Column Spaces Describe Linear Systems?
3. What Makes a Transformation Linear?
4. How Do Matrices Represent Linear Transformations?
5. How Does a Change of Basis Relate Two Matrix Representations?

How Do Coordinates Change with an Ordered Basis?

Coordinates with Respect to a Basis

- The standard basis for $\mathbb{R}^2$ is $\{e_1, e_2\}$. Any vector $\mathbf{x}$ in $\mathbb{R}^2$ can be expressed as a linear combination:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$$

- The scalars x_1 and x_2 are the coordinates of x with respect to the standard basis. For any basis $\{\mathbf{y}, \mathbf{z}\}$, a given vector $\mathbf{x}$ has a unique representation:

$$\mathbf{x} = \alpha \mathbf{y} + \beta \mathbf{z}.$$

- The scalars α and β are the coordinates of x with respect to the basis $\{\mathbf{y}, \mathbf{z}\}$.

Ordered Bases and Coordinate Vectors

- Let us order the basis elements so that $\mathbf{y}$ is the first basis vector and $\mathbf{z}$ is the second, and denote the ordered basis by $[\mathbf{y}, \mathbf{z}]$.

$$\mathbf{x} = \alpha\mathbf{y} + \beta\mathbf{z} \qquad \text{coordinate vector: } \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

- Reversing the order of the basis vectors also reverses the order of the coordinates:

$$\mathbf{x} = \beta\mathbf{z} + \alpha\mathbf{y} \qquad \text{in } [\mathbf{z}, \mathbf{y}]: \begin{bmatrix} \beta \\ \alpha \end{bmatrix}.$$

- Linear independence guarantees uniqueness. The order specifies which coefficient belongs to each basis vector.

Example 1: Coordinates in $\mathbb{R}^2$

$$\mathbf{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

- The determinant of the matrix with columns y and z is $2(4) - 1(1) = 7 \neq 0$, so **y** and **z** form a basis for $\mathbb{R}^2$.

$$3\mathbf{y} + \mathbf{z} = 3\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix} = \mathbf{x}.$$

- The coordinate vector is $(3, 1)^T$ with respect to $[\mathbf{y}, \mathbf{z}]$ and $(1, 3)^T$ with respect to $[\mathbf{z}, \mathbf{y}]$. The vector **x** is the same in both representations.

Example 1: Geometric Interpretation

- The coordinate vector specifies how to get from the origin to (7, 7) using the basis directions.
  - In [**y**, **z**], first use 3**y** and then **z**.
  - In [**z**, **y**], first use **z** and then 3**y**.

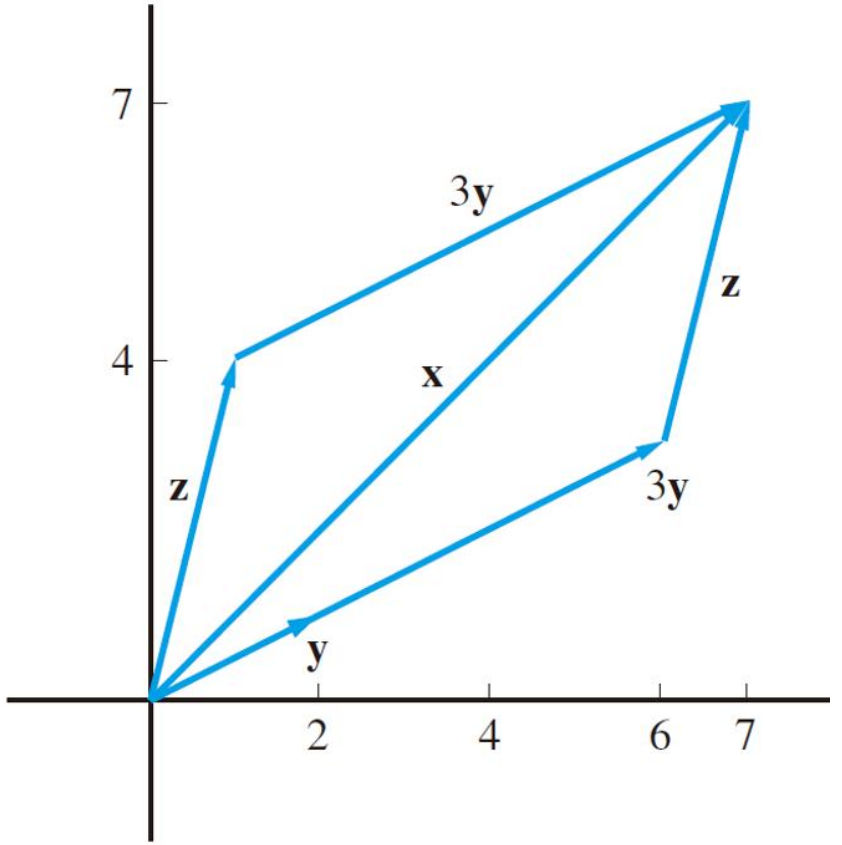


Figure 3.5.1.

Application 1: Population Migration

- The total population remains fixed. Each year, 6 percent of city residents move to the suburbs and 2 percent of suburban residents move to the city. Initially, 30 percent live in the city and 70 percent in the suburbs.

$$A = \begin{bmatrix} 0.94 & 0.02 \\ 0.06 & 0.98 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.30 \\ 0.70 \end{bmatrix}, \quad \mathbf{x}_n = A^n \mathbf{x}_0.$$

- The first coordinate gives the city proportion and the second the suburban proportion.
- Find the proportions after 10, 30, and 50 years and explain their long-term behavior.

Application 1: A Useful New Basis

- Choose the textbook basis vectors $\mathbf{u}_1 = (1, 3)^T$ and $\mathbf{u}_2 = (-1, 1)^T$. Multiplication by A has a simple effect on each:

$$A\mathbf{u}_1 = (0.94 + 0.06, 0.06 + 2.94)^T = (1, 3)^T = \mathbf{u}_1$$

$$A\mathbf{u}_2 = (-0.94 + 0.02, -0.06 + 0.98)^T$$

$$= (-0.92, 0.92)^T = 0.92\mathbf{u}_2$$

$$\mathbf{x}_0 = 0.25\mathbf{u}_1 - 0.05\mathbf{u}_2 = \begin{bmatrix} 0.25 + 0.05 \\ 0.75 - 0.05 \end{bmatrix} = \begin{bmatrix} 0.30 \\ 0.70 \end{bmatrix}.$$

Application 1: Long-Term Population

- Repeated multiplication leaves the $\mathbf{u}_1$ component unchanged and multiplies the u_2 component by 0.92 each year:

$$\mathbf{x}_n = 0.25\mathbf{u}_1 - 0.05(0.92)^n\mathbf{u}_2 = (0.25 + 0.05(0.92)^n,\ 0.75 - 0.05(0.92)^n)^T$$

$$\mathbf{x}_{10} \approx (0.27,\ 0.73)^T, \quad \mathbf{x}_{30} \approx (0.25,\ 0.75)^T, \quad \mathbf{x}_{50} \approx (0.25,\ 0.75)^T$$

- The displayed proportions are rounded to the nearest percent.
- As n increases, $(0.92)^n$ approaches 0, so $\mathbf{x}_n$ approaches the steady-state vector $(0.25,\ 0.75)^T$. Changing the basis makes this behavior clear.

Transition Matrices

- Section 3.5 continues with methods for changing coordinates between ordered bases.
- The transition matrix records how the old basis vectors are expressed in the new basis. Its inverse reverses the coordinate conversion.

Changing Coordinates: Two Problems

I. Given a vector $\mathbf{x} = (x_1, x_2)^T$, find its coordinates with respect to $\mathbf{u}_1$ and $\mathbf{u}_2$.

II. Given a vector $c_1\mathbf{u}_1 + c_2\mathbf{u}_2$, find its coordinates with respect to $\mathbf{e}_1$ and $\mathbf{e}_2$.

- We solve II first by expressing each old basis vector as a linear combination of the new basis vectors. This uses the matrix multiplication learned in Chapter 1.

The Transition Matrix to the Standard Basis

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \qquad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

$$\mathbf{u}_1 = 3\mathbf{e}_1 + 2\mathbf{e}_2, \qquad \mathbf{u}_2 = \mathbf{e}_1 + \mathbf{e}_2,$$
$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 = (3c_1 + c_2)\mathbf{e}_1 + (2c_1 + c_2)\mathbf{e}_2.$$

$$\mathbf{x} = \begin{bmatrix} 3c_1 + c_2 \\ 2c_1 + c_2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = U\mathbf{c}.$$

- U = ($\mathbf{u}_1$, $\mathbf{u}_2$) is the transition matrix from the ordered basis [$\mathbf{u}_1$, $\mathbf{u}_2$] to the standard basis [$\mathbf{e}_1$, $\mathbf{e}_2$]. Its columns are the old basis vectors expressed in the new basis.

The Reverse Transition Matrix

- The columns of U are linearly independent, so U is nonsingular. Multiplying $\mathbf{x} = U\mathbf{c}$ by U^{-1} gives the reverse conversion:

$$\mathbf{c} = U^{-1}\mathbf{x}, \qquad U^{-1} = \frac{1}{3(1)-1(2)}\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}.$$

- U^{-1} is the transition matrix from $[\mathbf{e}_1, \mathbf{e}_2]$ to $[\mathbf{u}_1, \mathbf{u}_2]$.
- The direction matters: U converts **c** to **x**, while U^{-1} converts **x** to **c**.

$$[\mathbf{u}_1, \mathbf{u}_2] \xrightarrow{\quad U \quad} [\mathbf{e}_1, \mathbf{e}_2]$$

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \qquad \mathbf{x} = U\mathbf{c} \ (\mathbf{c} = U^{-1}\mathbf{x}) \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 = x_1\mathbf{e}_1 + x_2\mathbf{e}_2$$

Example 2: Finding the Coordinates

Let $\mathbf{u}_1 = (3, 2)^T$, $\mathbf{u}_2 = (1, 1)^T$, and $\mathbf{x} = (7, 4)^T$. Find the coordinates of $\mathbf{x}$ with respect to $\mathbf{u}_1$ and $\mathbf{u}_2$.

$$\mathbf{c} = U^{-1}\mathbf{x} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 7-4 \\ -14+12 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

$$\mathbf{x} = 3\mathbf{u}_1 - 2\mathbf{u}_2 = 3\begin{bmatrix} 3 \\ 2 \end{bmatrix} - 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}.$$

- The coordinate vector is $(3, -2)^T$. Substituting the coordinates into the basis expansion verifies the answer.

Example 3: A Different Ordered Basis

Let $\mathbf{b}_1 = (1, -1)^T$ and $\mathbf{b}_2 = (-2, 3)^T$. Find the transition matrix from $[\mathbf{e}_1, \mathbf{e}_2]$ to $[\mathbf{b}_1, \mathbf{b}_2]$ and the coordinates of $\mathbf{x} = (1, 2)^T$.

$$B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}, \quad \det B = 3 - 2 = 1, \quad B^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}.$$

- B converts coordinates in $[\mathbf{b}_1, \mathbf{b}_2]$ to standard coordinates. Therefore, B^{-1} is the requested transition matrix.

$$\mathbf{c} = B^{-1}\mathbf{x} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3+4 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}.$$

- Thus, $\mathbf{x} = 7\mathbf{b}_1 + 3\mathbf{b}_2$. To verify, use the original basis vectors:

$$7\mathbf{b}_1 + 3\mathbf{b}_2 = 7 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + 3 \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 7-6 \\ -7+9 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Changing Between Two Arbitrary Bases

- Suppose **c** is the coordinate vector of **x** with respect to $[\mathbf{v}_1, \mathbf{v}_2]$. We seek its coordinate vector **d** with respect to $[\mathbf{u}_1, \mathbf{u}_2]$.

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = d_1\mathbf{u}_1 + d_2\mathbf{u}_2.$$

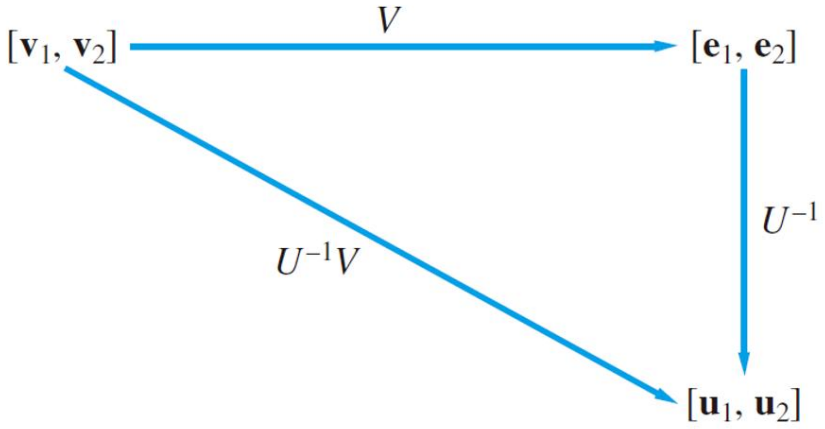


Figure 3.5.2.

- Set V = ($\mathbf{v}_1$, $\mathbf{v}_2$) and U = ($\mathbf{u}_1$, $\mathbf{u}_2$). Then

$$V\mathbf{c} = U\mathbf{d}, \qquad \mathbf{d} = U^{-1}V\mathbf{c}.$$

- The transition matrix from $[\mathbf{v}_1, \mathbf{v}_2]$ to $[\mathbf{u}_1, \mathbf{u}_2]$ is S = U^{-1}V. The input is **c** and the output is **d**.

$$[\mathbf{v}_1, \mathbf{v}_2] \xrightarrow{\quad S \quad} [\mathbf{u}_1, \mathbf{u}_2]$$

$$\mathbf{d} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \qquad \mathbf{c} = S\mathbf{d} \ (\mathbf{d} = S^{-1}\mathbf{c}) \qquad \mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$d_1\mathbf{v}_1 + d_2\mathbf{v}_2 = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$$

Example 4: A Transition Matrix

$$\mathbf{v}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 7 \\ 3 \end{bmatrix}, \quad \mathbf{u}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Find the transition matrix from $[\mathbf{v}_1, \mathbf{v}_2]$ to $[\mathbf{u}_1, \mathbf{u}_2]$.

$$S = U^{-1}V = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}$$
$$= \begin{bmatrix} 5-2 & 7-3 \\ -10+6 & -14+9 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -4 & -5 \end{bmatrix}.$$

Example 4: Meaning of the Columns

- The first column gives the coordinates of $\mathbf{v}_1$ in $[\mathbf{u}_1, \mathbf{u}_2]$. The second gives the coordinates of $\mathbf{v}_2$ in the same ordered basis.

$$\mathbf{v}_1 = 3\mathbf{u}_1 - 4\mathbf{u}_2 = \begin{bmatrix} 9-4 \\ 6-4 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix},$$

$$\mathbf{v}_2 = 4\mathbf{u}_1 - 5\mathbf{u}_2 = \begin{bmatrix} 12-5 \\ 8-5 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}.$$

- The conversion can be performed in two steps: $\mathbf{x} = V\mathbf{c}$ gives standard coordinates, then $\mathbf{d} = U^{-1}\mathbf{x}$ gives coordinates in $[\mathbf{u}_1, \mathbf{u}_2]$. Thus, $\mathbf{d} = U^{-1}V\mathbf{c}$.

Coordinate Vectors in a General Vector Space

Let V be a vector space and let $E = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be an ordered basis for V. If $\mathbf{v}$ is any element of V, then $\mathbf{v}$ can be written in the form

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$$

where $c_1, c_2, \ldots, c_n$ are scalars. Thus, we can associate with each vector v a unique vector $\mathbf{c} = (c_1, c_2, \ldots, c_n)^T$ in $\mathbb{R}^n$. The vector $\mathbf{c}$ defined in this way is called the coordinate vector of $\mathbf{v}$ with respect to the ordered basis E and is denoted $[\mathbf{v}]_E$. The c_i's are called the coordinates of $\mathbf{v}$ relative to E. The vector $\mathbf{v}$ belongs to V, while its coordinate vector $[\mathbf{v}]_E$ belongs to $\mathbb{R}^n$. The same definition applies to vectors, matrices, and polynomials.

Example 5: Two Ordered Bases for $\mathbb{R}^3$

$$E = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix} \right\},$$

$$F = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

$$\mathbf{x} = 3\mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3, \quad \mathbf{y} = \mathbf{v}_1 - 3\mathbf{v}_2 + 2\mathbf{v}_3$$

Find the transition matrix from E to F, and use it to find the coordinates of **x** and **y** with respect to F.

Example 5: Constructing U and V

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \qquad V = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 5 \\ 1 & 2 & 4 \end{bmatrix}.$$

- To compute U^{-1}, reduce $(U \mid I)$. Subtract row 1 from row 2, subtract row 3 from row 2, then subtract rows 2 and 3 from row 1.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$$

Example 5: The Transition Matrix

$$U^{-1}V = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 5 \\ 1 & 2 & 4 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 1 & -3 \\ -1 & -1 & 0 \\ 1 & 2 & 4 \end{bmatrix}.$$

- For example, the third column comes from $2(1) - 5 = -3$, $-1 + 5 - 4 = 0$, and 4.
- Each column gives the coordinates of the corresponding vector $\mathbf{v}_j$ in F.

Example 5: Coordinates of x and y

$$[\mathbf{x}]_F = \begin{bmatrix} 1 & 1 & -3 \\ -1 & -1 & 0 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3+2+3 \\ -3-2 \\ 3+4-4 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 3 \end{bmatrix},$$

$$[\mathbf{y}]_F = \begin{bmatrix} 1 & 1 & -3 \\ -1 & -1 & 0 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1-3-6 \\ -1+3 \\ 1-6+8 \end{bmatrix} = \begin{bmatrix} -8 \\ 2 \\ 3 \end{bmatrix}.$$

- Therefore, $\mathbf{x} = 8\mathbf{u}_1 - 5\mathbf{u}_2 + 3\mathbf{u}_3$ and $\mathbf{y} = -8\mathbf{u}_1 + 2\mathbf{u}_2 + 3\mathbf{u}_3$.

Example 5: Verification in Both Bases

$$3\mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3 = \begin{bmatrix} 3+4-1 \\ 3+6-5 \\ 3+4-4 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix},$$

$$8\mathbf{u}_1 - 5\mathbf{u}_2 + 3\mathbf{u}_3 = \begin{bmatrix} 8-5+3 \\ 8-10+6 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix}.$$

$$\mathbf{v}_1 - 3\mathbf{v}_2 + 2\mathbf{v}_3 = \begin{bmatrix} -3 \\ 2 \\ 3 \end{bmatrix} = -8\mathbf{u}_1 + 2\mathbf{u}_2 + 3\mathbf{u}_3.$$

- Both expansions describe the same vector. Only the coordinate vector changes.

Transition Matrices for a General Vector Space

Let E = $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ and F = $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be ordered bases for an n-dimensional vector space V. Express each old basis vector in the new basis:

$$\mathbf{w}_j = \sum_{i=1}^{n} s_{ij}\mathbf{v}_i, \qquad j = 1, \ldots, n.$$

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1n} \\ s_{21} & s_{22} & \cdots & s_{2n} \\ \vdots & \vdots & & \vdots \\ s_{n1} & s_{n2} & \cdots & s_{nn} \end{bmatrix}.$$

Column j of S contains the coordinates of $\mathbf{w}_j$ with respect to F. The row index i identifies the coefficient of $\mathbf{v}_i$.

Derivation of the Coordinate Conversion

- Let $\mathbf{x} = [\mathbf{v}]_E$ and $\mathbf{y} = [\mathbf{v}]_F$. Substitute the basis expansions for $\mathbf{w}_j$ and collect the coefficients of each $\mathbf{v}_i$:

$$\mathbf{v} = \sum_{j=1}^{n} x_j \mathbf{w}_j = \sum_{j=1}^{n} x_j \left(\sum_{i=1}^{n} s_{ij} \mathbf{v}_i \right)$$

$$= \sum_{i=1}^{n} \left(\sum_{j=1}^{n} s_{ij} x_j \right) \mathbf{v}_i.$$

$$y_i = \sum_{j=1}^{n} s_{ij} x_j \quad (i = 1, \ldots, n), \qquad \mathbf{y} = S\mathbf{x}.$$

- Uniqueness of coordinates in the basis F identifies these coefficients with the entries of $\mathbf{y}$.

Why a Transition Matrix Is Nonsingular

Suppose $S\mathbf{x} = 0$. The coordinate-conversion identity implies that

$$x_1\mathbf{w}_1 + \cdots + x_n\mathbf{w}_n = 0\mathbf{v}_1 + \cdots + 0\mathbf{v}_n = 0$$

Since $\mathbf{w}_1, \ldots, \mathbf{w}_n$ are linearly independent, $x_1 = \cdots = x_n = 0$. Thus, $S\mathbf{x} = 0$ has only the trivial solution, so S is nonsingular.

$$\mathbf{y} = S\mathbf{x} \quad \Longleftrightarrow \quad \mathbf{x} = S^{-1}\mathbf{y}.$$

S converts coordinates from E to F. Its inverse S^{-1} converts coordinates from F to E.

Example 6: Coordinates for Polynomials

In P_3, change from the ordered basis $[1, x, x^2]$ to $[1, 2x, 4x^2 - 2]$. In this textbook, P_3 consists of polynomials of degree at most 2, including the zero polynomial.

First express the second basis in the standard basis:

$$\begin{aligned} 1 &= 1 \cdot 1 + 0x + 0x^2, \\ 2x &= 0 \cdot 1 + 2x + 0x^2, \\ 4x^2 - 2 &= -2 \cdot 1 + 0x + 4x^2. \end{aligned}$$

The coefficient columns give the transition matrix from $[1, 2x, 4x^2 - 2]$ to $[1, x, x^2]$.

Example 6: The Two Transition Matrices

$$S = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \qquad S^{-1} = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}.$$

- S converts coordinates from $[1, 2x, 4x^2 - 2]$ to $[1, x, x^2]$. S^{-1} converts in the requested direction.
- For an independent check, if the new coordinates are $(\alpha, \beta, \gamma)^T$, expanding the polynomial gives standard coordinates $(\alpha - 2\gamma, 2\beta, 4\gamma)^T$. Solving these three equations gives the entries of S^{-1}.

Example 6: Coordinates and Verification

- Given any $p(x) = a + bx + cx^2$ in P_3, multiply the standard coordinate vector by S^{-1}:

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a + \frac{1}{2}c \\ \frac{1}{2}b \\ \frac{1}{4}c \end{bmatrix}.$$

$$\begin{aligned} p(x) &= (a + \tfrac{1}{2}c) \cdot 1 + (\tfrac{1}{2}b) \cdot 2x + \tfrac{1}{4}c(4x^2 - 2) \\ &= a + \tfrac{1}{2}c + bx + cx^2 - \tfrac{1}{2}c = a + bx + cx^2. \end{aligned}$$

Every Nonsingular Matrix Gives a Change of Basis

- Let S be an n × n nonsingular matrix and let $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be an ordered basis for V. Define $\mathbf{w}_j = \sum_i s_{ij}\mathbf{v}_i$, as before. Suppose a linear combination of the $\mathbf{w}_j$'s is zero:

$$\sum_{j=1}^{n} x_j \mathbf{w}_j = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} s_{ij} x_j \right) \mathbf{v}_i = \mathbf{0}.$$

- Independence of the $\mathbf{v}_i$'s implies $S\mathbf{x} = 0$. Since S is nonsingular, $\mathbf{x} = 0$. Therefore, the $\mathbf{w}_j$'s are linearly independent. There are n of them in an n-dimensional space, so they form a basis.
- S is the transition matrix from $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ to $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$.

How Do Row and Column Spaces Describe Linear Systems?

Row Vectors and Column Vectors

- If A is an m × n matrix, each row of A is an n-tuple of real numbers and hence can be considered as a vector in $\mathbb{R}^{1\times n}$. The m vectors corresponding to the rows of A will be referred to as the row vectors of A.
- Similarly, each column of A can be considered as a vector in $\mathbb{R}^m$, and we can associate n column vectors with the matrix A.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Row Space and Column Space

If A is an m × n matrix, the subspace of $\mathbb{R}^{1\times n}$ spanned by the row vectors of A is called the row space of A. The subspace of $\mathbb{R}^{m}$ spanned by the column vectors of A is called the column space of A.

- The row space contains linear combinations of rows.
- The column space contains linear combinations of columns.
- Their vectors generally belong to different spaces.

Example 1: Row Space and Column Space

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- The row space of A is the set of all 3-tuples of the form

$$\alpha(1,0,0) + \beta(0,1,0) = (\alpha, \beta, 0).$$

- The column space of A is the set of all vectors of the form

$$\alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

- Thus, the row space of A is a two-dimensional subspace of $\mathbb{R}^{1\times 3}$, and the column space of A is $\mathbb{R}^2$.

Theorem 3.6.1: Row Equivalent Matrices

Two row equivalent matrices have the same row space.

Proof:

If B is row equivalent to A, then B can be formed from A by a finite sequence of row operations. Every row of B is therefore a linear combination of the rows of A. Consequently, every linear combination of rows of B is also a linear combination of rows of A. The row space of B is a subspace of the row space of A. Every elementary row operation is reversible. Thus, A is row equivalent to B, and every row of A is a linear combination of the rows of B. The row space of A is therefore a subspace of the row space of B. Each row space contains the other, so the two row spaces are equal. This result explains why row reduction can identify a basis for the original row space.

Rank and a Basis for the Row Space

Definition:

The rank of a matrix A, denoted rank(A), is the dimension of the row space of A.

- To determine the rank of a matrix, we can reduce the matrix to row echelon form. The nonzero rows of the row echelon matrix will form a basis for the row space.
- By Theorem 3.6.1, this is also a basis for the row space of A. The rank equals the number of nonzero rows in its row echelon form.

Why the Nonzero Echelon Rows Form a Basis

- The nonzero rows span the row space because adding or removing zero rows does not change any linear combination.
- They are linearly independent. In a zero linear combination of the nonzero rows, the first leading column forces the coefficient of the first row to be zero. The next leading column forces the coefficient of the second row to be zero.
- Continuing through all leading columns forces every coefficient to be zero. Thus, the nonzero rows form a basis.

Example 2: Reduction to Row Echelon Form

$$A=\begin{bmatrix}1 & -2 & 3\\ 2 & -5 & 1\\ 1 & -4 & -7\end{bmatrix}\Rightarrow\begin{bmatrix}1 & -2 & 3\\ 0 & 1 & 5\\ 0 & 0 & 0\end{bmatrix}=U$$

- The nonzero rows (1, −2, 3) and (0, 1, 5) form a basis for the row space of U.
- Since U and A are row equivalent, they have the same row space. Thus, these same two row vectors form a basis for the row space of A.
- rank(A) = rank(U) = 2

Linear Systems and the Column Space

- Consider the system A$\mathbf{x}$ = $\mathbf{b}$:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\Rightarrow x_1 \underbrace{\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}}_{\mathbf{a}_1} + x_2 \underbrace{\begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}}_{\mathbf{a}_2} + \cdots + x_n \underbrace{\begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}}_{\mathbf{a}_n} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

- A solution provides coefficients that express $\mathbf{b}$ as a linear combination of the columns of A. Conversely, any such linear combination provides a solution of A$\mathbf{x}$ = $\mathbf{b}$.
- A$\mathbf{x}$ = $\mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A

Theorem 3.6.2: Consistency Theorem for Linear Systems

A linear system $Ax = b$ is consistent if and only if b is in the column space of A.

Proof:

If $A\mathbf{x} = \mathbf{b}$ is consistent, a solution gives coefficients $x_1, \ldots, x_n$ with $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$. Hence, $\mathbf{b}$ is in the column space.

If $\mathbf{b}$ is in the column space, it has such an expression. The vector $\mathbf{x} = (x_1, \ldots, x_n)^T$ then satisfies $A\mathbf{x} = \mathbf{b}$, so the system is consistent.

Homogeneous Systems and Linear Independence

- Replacing b by the zero vector gives

$$A\mathbf{x} = \mathbf{0} \quad \Longleftrightarrow \quad x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}.$$

- The system A**x** = 0 will have only the trivial solution **x** = **0** if and only if the column vectors of A are linearly independent.
- This follows directly from the definition of linear independence:

  a zero linear combination must have all coefficients equal to zero.

Theorem 3.6.3: Existence and Uniqueness

Let A be an $m \times n$ matrix. The linear system $A\mathbf{x} = \mathbf{b}$ is consistent for every $b \in \mathbb{R}^m$ if and only if the column vectors of A span $\mathbb{R}^m$. The system $A\mathbf{x} = \mathbf{b}$ has at most one solution for every $\mathbf{b} \in \mathbb{R}^m$ if and only if the column vectors of A are linearly independent.

- "At most one" allows either no solution or one solution. Existence for every **b** requires the spanning condition in the first statement.

Theorem 3.6.3: Proof of the Existence Statement

- By Theorem 3.6.2, A$\mathbf{x} = \mathbf{b}$ is consistent exactly when b belongs to the column space of A.
- Therefore, A$\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$ exactly when the column space of A is all of $\mathbb{R}^m$.
- By the definition of column space, this is exactly the statement that the column vectors of A span $\mathbb{R}^m$.

Theorem 3.6.3: Proof of the Uniqueness Statement

- If A$\mathbf{x} = \mathbf{b}$ has at most one solution for every $\mathbf{b}$, then A$\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- Hence, the columns of A are linearly independent.
- Conversely, suppose the columns are linearly independent. Then A$\mathbf{x} = \mathbf{0}$ has only the trivial solution. If $\mathbf{x}_1$ and $\mathbf{x}_2$ both satisfy A$\mathbf{x}$ = b, then

$$A(\mathbf{x}_1 - \mathbf{x}_2) = A\mathbf{x}_1 - A\mathbf{x}_2 = \mathbf{b} - \mathbf{b} = \mathbf{0}.$$

- It follows that $\mathbf{x}_1 - \mathbf{x}_2 = 0$, so $\mathbf{x}_1 = \mathbf{x}_2$. Thus, there is at most one solution.

The Number of Columns and the Dimension

- Let A be an $m \times n$ matrix. Its n column vectors belong to $\mathbb{R}^m$.
- If the column vectors span $\mathbb{R}^m$, then $n \geq m$, because no set of fewer than m vectors can span $\mathbb{R}^m$.
- If the column vectors are linearly independent, then $n \leq m$, because every set of more than m vectors in $\mathbb{R}^m$ is linearly dependent.
- Thus, if the column vectors of A form a basis for $\mathbb{R}^m$, then $n = m$.

Corollary 3.6.4: Nonsingularity and a Column Basis

An $n \times n$ matrix A is nonsingular if and only if the column vectors of A form a basis for $\mathbb{R}^n$.

Proof:

If A is nonsingular, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Therefore, its n columns are linearly independent and form a basis for $\mathbb{R}^n$.

Conversely, if the columns form a basis, they are linearly independent. Thus, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Since A is square, A is nonsingular.

Nullity and the Rank–Nullity Theorem

- The dimension of the null space of a matrix is called the nullity of the matrix.

$$\text{nullity of } A = \dim N(A).$$

- Theorem 3.6.5: The Rank-Nullity Theorem.

  If A is an m × n matrix, then the rank of A plus the nullity of A equals n.

$$\text{rank}(A) + \dim N(A) = n.$$

  The total is the number of columns of A, which is the number of unknowns in $A\mathbf{x} = \mathbf{0}$.

Theorem 3.6.5: Lead Variables and Free Variables

Proof:

Let U be the reduced row echelon form of A. The system A$\mathbf{x} = \mathbf{0}$ is equivalent to U$\mathbf{x} = \mathbf{0}$.

If A has rank r, then U has r nonzero rows. Consequently, U$\mathbf{x} = \mathbf{0}$ has r lead variables and n − r free variables.

The dimension of N(A) equals the number of free variables.

Therefore, $\dim N(A) = n - r, \qquad \text{rank}(A) + \dim N(A) = r + (n - r) = n.$

Why Free Variables Determine the Null Space Dimension

- In the reduced system U$\mathbf{x} = \mathbf{0}$, solve each lead variable in terms of the free variables.
- Set one free variable equal to 1 and all other free variables equal to 0. Repeat this for each of the $n - r$ free variables to obtain $n - r$ solution vectors.
- These vectors span N(A), because any choice of free variables gives the corresponding linear combination. They are linearly independent, because their coordinates corresponding to free variables force every coefficient in a zero combination to vanish.

Example 3: A Row Space Basis and a Null Space Basis

Find a basis for the row space of A and a basis for N(A). Verify that dim N(A) = n − r.

$$A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & 1 & 5 \end{bmatrix}$$

- Subtract twice row 1 from row 2, and subtract row 1 from row 3. Multiply row 2 by −1. Then subtract twice row 2 from row 3, and add row 2 to row 1.

$$U = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Thus, {(1, 2, 0, 3), (0, 0, 1, 2)} is a basis for the row space of A, and rank(A) = 2.
- The lead variables are x_1 and x_3. The free variables are x_2 and x_4.

Example 3: Description of the Null Space

- Because A**x** = **0** and U**x** = **0** are equivalent, **x** is in N(A) exactly when

$$x_1 + 2x_2 + 3x_4 = 0, \qquad x_3 + 2x_4 = 0.$$

- Solve for the lead variables and set $x_2 = \alpha$ and $x_4 = \beta$.

$$x_1 = -2\alpha - 3\beta, \qquad x_3 = -2\beta.$$

$$\mathbf{x} = \begin{bmatrix} -2\alpha - 3\beta \\ \alpha \\ -2\beta \\ \beta \end{bmatrix} = \alpha \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -3 \\ 0 \\ -2 \\ 1 \end{bmatrix}.$$

Example 3: Basis Verification and Nullity

- The displayed expression shows that $(-2, 1, 0, 0)^T$ and $(-3, 0, -2, 1)^T$ span N(A).
- They are linearly independent: coordinate 2 of a zero linear combination gives the first coefficient as zero, and coordinate 4 gives the second coefficient as zero.

$$A\begin{bmatrix}-2\\1\\0\\0\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix},\qquad A\begin{bmatrix}-3\\0\\-2\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}.$$

$$\dim N(A)=2=4-2=n-r.$$

The Column Space and Dependency Relations

- For the matrix U in Example 3, the columns $\mathbf{u}_1$ and $\mathbf{u}_3$ are linearly independent, while

$$\mathbf{u}_2 = 2\mathbf{u}_1, \qquad \mathbf{u}_4 = 3\mathbf{u}_1 + 2\mathbf{u}_3.$$

- The same relations hold for the columns of A. The columns $\mathbf{a}_1$ and $\mathbf{a}_3$ are linearly independent, while

$$\mathbf{a}_2 = 2\mathbf{a}_1, \qquad \mathbf{a}_4 = 3\mathbf{a}_1 + 2\mathbf{a}_3.$$

- In general, row reduction preserves these dependency relations because $A\mathbf{x} = \mathbf{0}$ if and only if $U\mathbf{x} = \mathbf{0}$. It need not preserve the column space itself.

Theorem 3.6.6: Equality of the Two Dimensions

If A is an $m \times n$ matrix, the dimension of the row space of A equals the dimension of the column space of A.

- rank(A) is also the dimension of the column space. The two spaces need not be the same: the row space lies in $\mathbb{R}^{1\times n}$ and the column space lies in $\mathbb{R}^m$.
- The proof first identifies independent original columns using leading 1's in an echelon form. It then applies the same argument to A^T.

Theorem 3.6.6: Independent Original Columns

Proof:

Suppose rank(A) = r. Its row echelon form U has r leading 1's. The corresponding r columns of U are linearly independent.

Delete the columns corresponding to free variables from U to obtain U_L. Delete the same columns from A to obtain A_L. Then A_L and U_L are row equivalent.

If $A_L\mathbf{x} = \mathbf{0}$, then $U_L\mathbf{x} = \mathbf{0}$. Since the columns of U_L are linearly independent, $\mathbf{x} = \mathbf{0}$.

Thus the r columns of A_L are linearly independent.

$$\dim(\text{column space of } A) \geq r = \dim(\text{row space of } A).$$

Theorem 3.6.6: The Transpose Argument

The preceding argument applies to any matrix. Apply it to A^T to obtain

$$\begin{aligned}\dim(\text{row space of } A) &= \dim(\text{column space of } A^T) \\ &\geq \dim(\text{row space of } A^T) \\ &= \dim(\text{column space of } A).\end{aligned}$$

The previous slide gave the reverse inequality. Therefore the two dimensions are equal.

This completes the proof.

A Basis for the Column Space

- Use the row echelon form U of A to identify the columns containing leading 1's. Take the corresponding columns from the original matrix A.
- Theorem 3.6.6 shows that these rank(A) independent columns form a basis for the column space of A.
- The row echelon form U tells us only which columns of A to use. We cannot generally use the column vectors from U, since U and A may have different column spaces.

Example 4: Selection of Original Columns

Find a basis for the column space of

$$A = \begin{bmatrix} 1 & -2 & 1 & 1 & 2 \\ -1 & 3 & 0 & 2 & -2 \\ 0 & 1 & 1 & 3 & 4 \\ 1 & 2 & 5 & 13 & 5 \end{bmatrix} = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5).$$

- The row echelon form of A is

$$U = \begin{bmatrix} 1 & -2 & 1 & 1 & 2 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 4: The Column Space Basis

- The leading 1's occur in columns 1, 2, and 5. Therefore the following columns of the original matrix A form a basis for its column space:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \qquad \mathbf{a}_2 = \begin{bmatrix} -2 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \qquad \mathbf{a}_5 = \begin{bmatrix} 2 \\ -2 \\ 4 \\ 5 \end{bmatrix}.$$

$$\text{rank}(A) = 3.$$

- The three nonzero rows of U form a basis for the row space of A. With five columns, A has nullity $5 - 3 = 2$.

Example 5: Dimension of a Spanned Subspace

Find the dimension of the subspace of $\mathbb{R}^4$ spanned by

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 5 \\ -3 \\ 2 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 4 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 3 \\ 8 \\ -5 \\ 4 \end{bmatrix}.$$

- The subspace Span(x_1, x_2, x_3, x_4) is the column space of the matrix

$$X = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 2 & 5 & 4 & 8 \\ -1 & -3 & -2 & -5 \\ 0 & 2 & 0 & 4 \end{bmatrix}.$$

Example 5: Reduction and Dimension

- The row echelon form of X is

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- The leading 1's occur in columns 1 and 2, so $\mathbf{x}_1$ and $\mathbf{x}_2$ form a basis for the column space of X.

$$\dim \text{Span}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4) = \text{rank}(X) = 2.$$

Example 5: Verification of the Spanning Relations

- The other two vectors are linear combinations of the selected basis vectors:

$$\mathbf{x}_3 = 2\mathbf{x}_1, \qquad \mathbf{x}_4 = -\mathbf{x}_1 + 2\mathbf{x}_2.$$

$$-\begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 2 \\ 5 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \\ -5 \\ 4 \end{bmatrix}.$$

- The vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ are linearly independent: coordinate 4 of a zero linear combination forces the coefficient of $\mathbf{x}_2$ to vanish, and coordinate 1 then forces the coefficient of $\mathbf{x}_1$ to vanish. Thus, the dimension is 2.

What Makes a Transformation Linear?

Definition of a Linear Transformation

A mapping L from a vector space V into a vector Space W is said to be a **linear transformation** if

$$L(\alpha \mathbf{v}_1 + \beta \mathbf{v}_2) = \alpha L(\mathbf{v}_1) + \beta L(\mathbf{v}_2)$$

for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and for all scalars α and β.

- $L : V \rightarrow W$. When V and W are the same, a linear transformation L: V $\rightarrow$ V is called a linear operator on V.

Equivalent Conditions for Linearity

- The defining condition gives the following two conditions.

$$\begin{aligned} L(\mathbf{v}_1+\mathbf{v}_2) &= L(\mathbf{v}_1)+L(\mathbf{v}_2) \quad (\alpha=\beta=1), \\ L(\alpha\mathbf{v}) &= \alpha L(\mathbf{v}) \qquad\qquad\quad (\beta=0). \end{aligned}$$

- Conversely, if both conditions hold, then

$$\begin{aligned} L(\alpha\mathbf{v}_1+\beta\mathbf{v}_2) &= L(\alpha\mathbf{v}_1)+L(\beta\mathbf{v}_2) \\ &= \alpha L(\mathbf{v}_1)+\beta L(\mathbf{v}_2). \end{aligned}$$

- Thus, a mapping is a linear transformation if and only if it satisfies both conditions for all vectors and scalars.

Example 1: Stretching in $\mathbb{R}^2$

- Let L be the operator defined by $L(\mathbf{x}) = 3\mathbf{x}$ for each $\mathbf{x} \in \mathbb{R}^2$.

$$L(\alpha\mathbf{x}) = 3(\alpha\mathbf{x}) = \alpha(3\mathbf{x}) = \alpha L(\mathbf{x})$$

$$\begin{aligned} L(\mathbf{x}+\mathbf{y}) &= 3(\mathbf{x}+\mathbf{y}) \\ &= 3\mathbf{x}+3\mathbf{y} = L(\mathbf{x})+L(\mathbf{y}). \end{aligned}$$

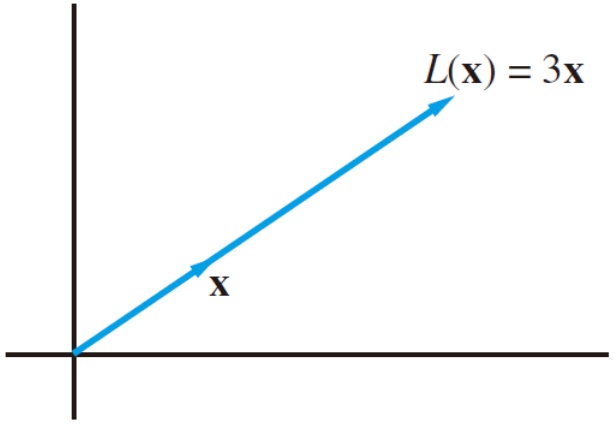


Figure 4.1.1.

- Therefore, L is a linear operator. It stretches each vector by a factor of 3, as in Figure 4.1.1.
- More generally, for a positive scalar α, the linear operator $F(\mathbf{x}) = \alpha\mathbf{x}$ stretches or shrinks vectors by a factor of α.

Example 2: Projection onto the x_1-Axis

- Consider the mapping $L(\mathbf{x}) = x_1\mathbf{e}_1$ for each $\mathbf{x} = (x_1, x_2)^T \in \mathbb{R}^2$. Thus $L(\mathbf{x}) = (x_1, 0)^T$.

  If $\mathbf{y} = (y_1, y_2)^T$, then

$$L(\alpha\mathbf{x} + \beta\mathbf{y}) = L\left(\alpha\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \beta\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = \boldsymbol{L}\left(\begin{bmatrix} \alpha x_1 + \beta y_1 \\ \alpha x_2 + \beta y_2 \end{bmatrix}\right) = \begin{bmatrix} \alpha x_1 + \beta y_1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} \beta y_1 \\ 0 \end{bmatrix} = \alpha\begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \beta\begin{bmatrix} y_1 \\ 0 \end{bmatrix} = \alpha L(\mathbf{x}) + \beta L(\mathbf{y})$$

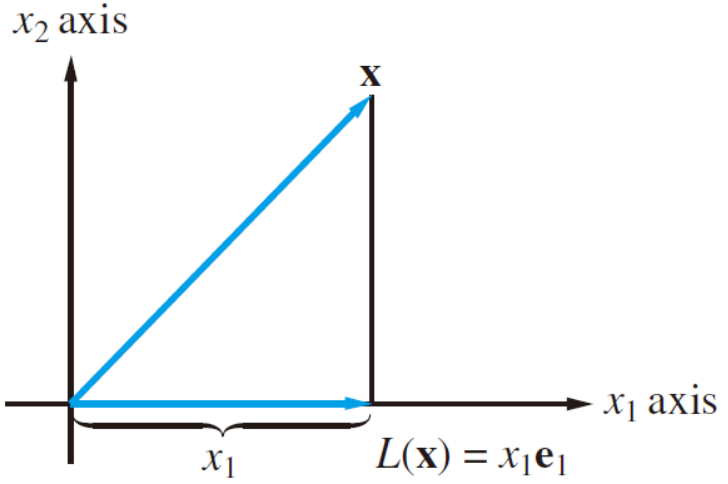


Figure 4.1.2.

- Hence, L is a linear operator. It projects a vector onto the x_1-axis by keeping x_1 and setting the second coordinate to zero. See Figure 4.1.2.

Example 3: Reflection about the x_1-Axis

- Let L be the operator defined by $L(\mathbf{x}) = (x_1, -x_2)^T$ for each $\mathbf{x} = (x_1, x_2)^T$ in $\mathbb{R}^2$.
- For each $\mathbf{x} = (x_1, x_2)^T$ and $\mathbf{y} = (y_1, y_2)^T$

$$L(\alpha\mathbf{x}+\beta\mathbf{y}) = L\left(\alpha\begin{bmatrix}x_1\\x_2\end{bmatrix}+\beta\begin{bmatrix}y_1\\y_2\end{bmatrix}\right) = \boldsymbol{L}\left(\begin{bmatrix}\alpha x_1+\beta y_1\\ \alpha x_2+\beta y_2\end{bmatrix}\right) = \begin{bmatrix}\alpha x_1+\beta y_1\\ -(\alpha x_2+\beta y_2)\end{bmatrix}$$

$$= \begin{bmatrix}\alpha x_1\\ -\alpha x_2\end{bmatrix}+\begin{bmatrix}\beta y_1\\ -\beta y_2\end{bmatrix} = \alpha\begin{bmatrix}x_1\\ -x_2\end{bmatrix}+\beta\begin{bmatrix}y_1\\ -y_2\end{bmatrix} = \alpha L(\mathbf{x}) + \beta L(\mathbf{y})$$

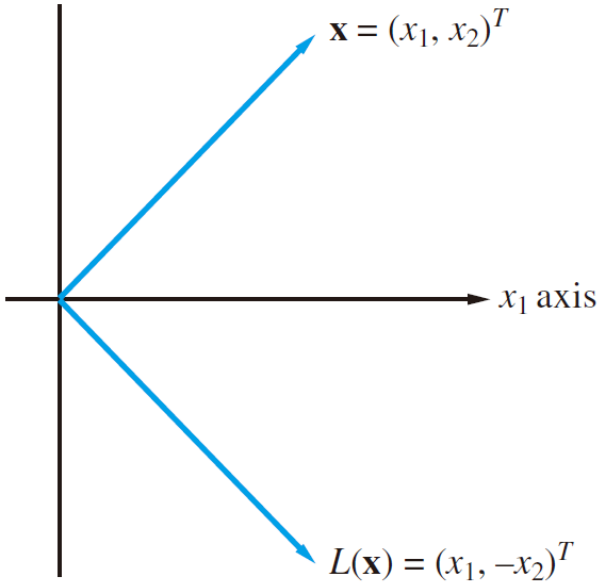


Figure 4.1.3.

- Therefore, L is a linear operator. It reflects each vector about the x_1-axis. The first coordinate stays the same and the second coordinate changes sign. See Figure 4.1.3.

Example 4: Counterclockwise Rotation by 90°

- Let L be the operator defined by $L(\mathbf{x}) = (-x_2, x_1)^T$ for each $\mathbf{x} = (x_1, x_2)^T$ in $\mathbb{R}^2$.
- For each $\mathbf{x} = (x_1, x_2)^T$ and $\mathbf{y} = (y_1, y_2)^T$

$$L(\alpha\mathbf{x}+\beta\mathbf{y}) = L\left(\alpha\begin{bmatrix}x_1\\x_2\end{bmatrix}+\beta\begin{bmatrix}y_1\\y_2\end{bmatrix}\right) = L\left(\begin{bmatrix}\alpha x_1+\beta y_1\\ \alpha x_2+\beta y_2\end{bmatrix}\right) = \begin{bmatrix}-(\alpha x_2+\beta y_2)\\ \alpha x_1+\beta y_1\end{bmatrix}$$

$$= \begin{bmatrix}-\alpha x_2\\ \alpha x_1\end{bmatrix}+\begin{bmatrix}-\beta y_2\\ \beta y_1\end{bmatrix} = \alpha\begin{bmatrix}-x_2\\ x_1\end{bmatrix}+\beta\begin{bmatrix}-y_2\\ y_1\end{bmatrix} = \alpha L(\mathbf{x}) + \beta L(\mathbf{y})$$

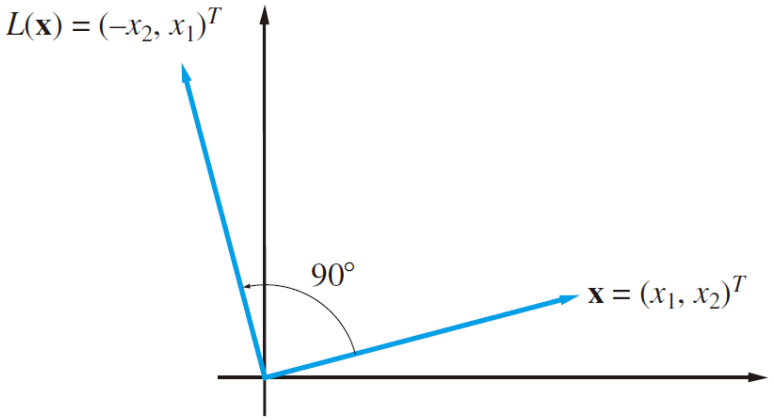


Figure 4.1.4.

- The operator rotates each vector in R² by 90° in the counterclockwise direction.

  See Figure 4.1.4.

Example 5: A Transformation from $\mathbb{R}^2$ to $\mathbb{R}^1$

- The mapping L: $\mathbb{R}^2 \rightarrow \mathbb{R}^1$ defined by $L(\mathbf{x}) = x_1 + x_2$ is a linear transformation.

$$\begin{aligned} L(\alpha\mathbf{x}+\beta\mathbf{y}) &= (\alpha x_1+\beta y_1)+(\alpha x_2+\beta y_2) \\ &= \alpha(x_1+x_2)+\beta(y_1+y_2) \\ &= \alpha L(\mathbf{x})+\beta L(\mathbf{y}). \end{aligned}$$

- The input has two coordinates and the output is a real number. The domain and codomain of a linear transformation can be different vector spaces.

Example 6: A Mapping That Is Not Linear

- Consider M: $\mathbb{R}^2 \rightarrow \mathbb{R}^1$ defined by $M(\mathbf{x}) = (x_1^2 + x_2^2)^{1/2}$.

$$M(\alpha x) = M\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = M\left(\begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix}\right) = \left[(\alpha x_1)^2 + (\alpha x_2)^2\right]^{1/2}$$

$$= |\alpha| \ (x_1^2 + x_2^2)^{1/2} = |\alpha| M(x)$$

$$\alpha M(x) = \alpha \ (x_1^2 + x_2^2)^{1/2}$$

- If $\alpha < 0$ and $x \neq 0$, then $|\alpha|M(\mathbf{x})$ differs from $\alpha M(\mathbf{x})$. Thus $M(\alpha\mathbf{x}) = \alpha M(\mathbf{x})$ fails, and M is not a linear transformation.
- Linearity must hold for every scalar, including negative scalars.

Example 7: A Transformation from $\mathbb{R}^2$ to $\mathbb{R}^3$

- Let L: $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $L(\mathbf{x}) = (x_2, x_1, x_1 + x_2)^T$.

$$L(\alpha x) = L\left(\alpha\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = L\left(\begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix}\right) = \begin{bmatrix} \alpha x_2 \\ \alpha x_1 \\ \alpha x_1 + \alpha x_2 \end{bmatrix} = \alpha\begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix} = \alpha L(x)$$

$$L(\mathbf{x}+\mathbf{y}) = L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = L\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}\right)$$

$$= \begin{bmatrix} x_2 + y_2 \\ x_1 + y_1 \\ x_1 + y_1 + x_2 + y_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix} + \begin{bmatrix} y_2 \\ y_1 \\ y_1 + y_2 \end{bmatrix} = L(\mathbf{x}) + L(\mathbf{y})$$

- Both conditions hold. Therefore, L is a linear transformation.

Example 7: The Matrix Form

- The transformation in Example 7 can also be written using a matrix.

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$A\mathbf{x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix} = L(\mathbf{x})$$

- The matrix has two columns because the input belongs to $\mathbb{R}^2$ and three rows because the output belongs to $\mathbb{R}^3$.

Matrices Define Linear Transformations

- If A is any m × n matrix, define the transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ by

$$L_A(\mathbf{x}) = A\mathbf{x} \qquad \text{for each } \mathbf{x} \in \mathbb{R}^n.$$

$$\begin{aligned} L_A(\alpha\mathbf{x}+\beta\mathbf{y}) &= A(\alpha\mathbf{x}+\beta\mathbf{y}) \\ &= \alpha A\mathbf{x}+\beta A\mathbf{y} \\ &= \alpha L_A(\mathbf{x})+\beta L_A(\mathbf{y}). \end{aligned}$$

- Thus, each m × n matrix defines a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. This extends the matrix multiplication studied in Chapter 1.

Zero Vectors and Additive Inverses

- Let L: V $\rightarrow$ W be a linear transformation. The symbols 0_V and 0_w denote the zero vectors in V and W, respectively.

$$L(\mathbf{0}_V) = L(0\mathbf{v}) = 0L(\mathbf{v}) = \mathbf{0}_W.$$

- For each v $\in$ V, linearity also gives

$$\mathbf{0}_W = L(\mathbf{0}_V) = L(\mathbf{v} + (-\mathbf{v})) = L(\mathbf{v}) + L(-\mathbf{v}).$$

- Therefore, $L(-\mathrm{v})$ is the additive inverse of $L(\mathrm{v})$, so

$$L(-\mathbf{v}) = -L(\mathbf{v}).$$

Linear Combinations under a Transformation

- For vectors $v_1, \ldots, v_n \in V$ and scalars $\alpha_1, \ldots, \alpha_n$, a linear transformation satisfies

$$L\left(\sum_{i=1}^{n} \alpha_i \mathbf{v}_i\right) = \sum_{i=1}^{n} \alpha_i L(\mathbf{v}_i).$$

- For n = 1 this is the scalar multiplication condition. Suppose the formula holds for n = k. Then

$$\begin{aligned} L\left(\sum_{i=1}^{k+1} \alpha_i \mathbf{v}_i\right) &= L\left(\sum_{i=1}^{k} \alpha_i \mathbf{v}_i\right) + L(\alpha_{k+1}\mathbf{v}_{k+1}) \\ &= \sum_{i=1}^{k} \alpha_i L(\mathbf{v}_i) + \alpha_{k+1} L(\mathbf{v}_{k+1}). \end{aligned}$$

- Mathematical induction proves the statement for every positive integer n.

Example 8: The Identity Operator

- If V is any vector space, the identity operator I is defined by

$$I(\mathbf{v}) = \mathbf{v} \quad \text{for all } \mathbf{v} \in V$$

$$I(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2$$

$$= \alpha I(\mathbf{v}_1) + \beta I(\mathbf{v}_2)$$

- Therefore, I is a linear transformation that maps V into itself.

  Each vector remains unchanged.

Example 9: Integration as a Linear Transformation

- Let L be the mapping from C[a, b] to $\mathbb{R}^1$ defined by

$$L(f) = \int_a^b f(x)\,dx.$$

- Here C[a, b] is the vector space of continuous real-valued functions on [a, b].

For $f, g \in$ C[a, b],

$$\begin{aligned} L(\alpha f + \beta g) &= \int_a^b (\alpha f + \beta g)(x)\,dx \\ &= \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx \\ &= \alpha L(f) + \beta L(g). \end{aligned}$$

- Thus, L is a linear transformation. Its input is a function and its output is a real number.

Example 10: Differentiation as a Linear Transformation

- Let D map $C^1[a, b]$ into $C[a, b]$, with $D(f) = f'$ (the derivative of f).
- $C^1[a, b]$ consists of functions with a continuous first derivative on [a, b].

  Thus, the derivative belongs to C[a, b].
- $C^1[a, b]$ consists of functions with a continuous first derivative on [a, b]. Thus the derivative belongs to C[a, b].

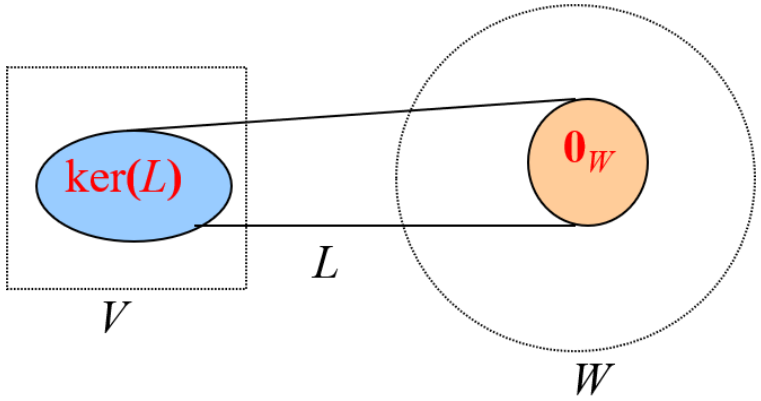
$$\begin{aligned} D(\alpha f + \beta g) &= (\alpha f + \beta g)' \\ &= \alpha f' + \beta g' \\ &= \alpha D(f) + \beta D(g). \end{aligned}$$

- Differentiation is a linear transformation between these function spaces.

Definition of the Kernel

Let L: $V \to W$ be a linear transformation. The **kernel** of L, denoted $\ker(L)$, is define by

$$\ker(L) = \{\mathbf{v} \in V \mid L(\mathbf{v}) = \mathbf{0}_W\}$$



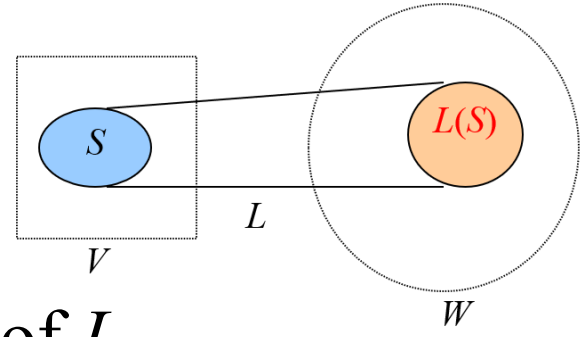
- The kernel contains the input vectors that L maps to the zero vector of W.
- For the transformation defined by A, the kernel is the null space from §3.6:

$$\ker(L_A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\} = N(A).$$

Definition of Image and Range

Let $L: V \to W$ be a linear transformation and let S be a subspace of V. The **image** of S, denoted $L(S)$, is defined by

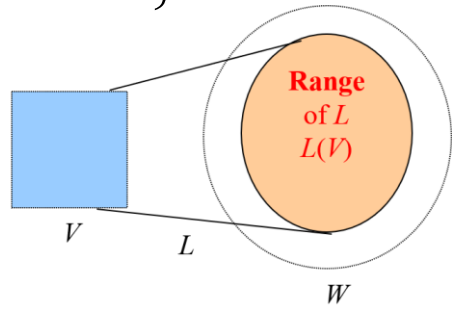
$$L(S) = \{\mathbf{w} \in W \mid \mathbf{w} = L(\mathbf{v}) \text{ for some } \mathbf{v} \in S\}$$



The **image** of the entire vector space, $L(V)$, is called the **range** of L.

- For the transformation defined by A, the range is the column space of A, since Ax is a linear combination of the columns of A.

$$L_A(\mathbb{R}^n) = \text{column space of } A.$$



Theorem 4.1.1

If $L: V \to W$ is a linear transformation and S is a subspace of V, then

(1) $\ker(L)$ is a subspace of V

(2) $L(S)$ is a subspace of W

- In particular, the range $L(V)$ is a subspace of W.
- For each part, the proof checks that the set is nonempty and is closed under scalar multiplication and vector addition.

Proof of Theorem 4.1.1: The Kernel

- Since $L(\mathbf{0}_V) = \mathbf{0}w$, the vector $\mathbf{0}_V$ belongs to ker(L). Thus, ker(L) is nonempty.
- If $\mathbf{v} \in \ker(L)$ and α is a scalar, then

$$L(\alpha\mathbf{v}) = \alpha L(\mathbf{v}) = \alpha\mathbf{0}_W = \mathbf{0}_W.$$

- Therefore, $\alpha\mathbf{v} \in \ker(L)$. If $\mathbf{v}_1, \mathbf{v}_2 \in \ker(L)$, then

$$L(\mathbf{v}_1 + \mathbf{v}_2) = L(\mathbf{v}_1) + L(\mathbf{v}_2) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W.$$

- Thus, $\mathbf{v}_1 + \mathbf{v}_2 \in \ker(L)$. Both closure conditions hold, so ker(L) is a subspace of V.

Proof of Theorem 4.1.1: The Image

- Since S is a subspace, $\mathbf{0}_v \in$ S. Hence $\mathbf{0}w = L(\mathbf{0}_v) \in L(\text{S})$, so $L(\text{S})$ is nonempty.
- If $\mathbf{w} \in L(\text{S})$, then $\mathbf{w} = L(\text{v})$ for some $\mathbf{v} \in$ S. For any scalar α,

$$\alpha\mathbf{w} = \alpha L(\mathbf{v}) = L(\alpha\mathbf{v}) \in L(S),$$

because αv $\in$ S. If $\mathbf{w}_1 = L(\mathbf{v}_1)$ and $\mathbf{w}_2 = L(\mathbf{v}_2)$ with $\mathbf{v}_1, \mathbf{v}_2 \in$ S, then

$$\mathbf{w}_1 + \mathbf{w}_2 = L(\mathbf{v}_1) + L(\mathbf{v}_2) = L(\mathbf{v}_1 + \mathbf{v}_2) \in L(S),$$

because $\mathbf{v}_1 + \mathbf{v}_2 \in$ S. Therefore, $L(\text{S})$ is a subspace of W.

Example 11: Kernel of a Projection

- Let L be the linear operator on $\mathbb{R}^2$ defined by

$$L(\mathbf{x}) = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}.$$

- A vector x belongs to ker(L) if and only if $x_1 = 0$. The second coordinate is free.

$$\ker(L) = \left\{ \begin{bmatrix} 0 \\ x_2 \end{bmatrix} : x_2 \in \mathbb{R} \right\} = \text{Span}(\mathbf{e}_2).$$

- Thus, ker(L) is the one-dimensional subspace of $\mathbb{R}^2$ spanned by e_2.

  It is the x_2-axis.

Example 11: Range of a Projection

- Every output of $L(x) = (x_1, 0)^T$ is a multiple of e_1. Conversely, any multiple of e_1 is an output, since $L((a, 0)^T) = (a, 0)^T$.

$$L(\mathbb{R}^2) = \left\{ \begin{bmatrix} a \\ 0 \end{bmatrix} : a \in \mathbb{R} \right\} = \text{Span}(\mathbf{e}_1).$$

- Hence, the range of L is the one-dimensional subspace spanned by e_1, the x_1-axis.
- The kernel describes inputs sent to zero. The range describes all possible outputs.

Example 12: Determining the Kernel

- Let L: $\mathbb{R}^3 \to \mathbb{R}^2$ be defined by $L(\mathbf{x}) = (x_1 + x_2, x_2 + x_3)^T$.

  Let S be the subspace of $\mathbb{R}^3$ spanned by e_1 and e_3.

- If $\mathbf{x} \in \ker(L)$, then $x_1 + x_2 = 0, \qquad x_2 + x_3 = 0.$

- Set the free variable $x_3 = a$. Then $x_2 = -a$ and $x_1 = a$, so

$$\mathbf{x} = a \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \qquad \ker(L) = \text{Span}\left(\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right).$$

- Therefore, ker(L) is a one-dimensional subspace of $\mathbb{R}^3$.

Example 12: The Image of S and the Range

- Since S is spanned by e_1 and e_3, each vector in S has the form

$$\mathbf{x} = a\mathbf{e}_1 + b\mathbf{e}_3 = \begin{bmatrix} a \\ 0 \\ b \end{bmatrix}.$$

$$L(\mathbf{x}) = \begin{bmatrix} a+0 \\ 0+b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

- The scalars a and b are arbitrary. Thus, every vector in $\mathbb{R}^2$ occurs as the image of a vector in S.

$$L(S) = \mathbb{R}^2.$$

- Because $S \subseteq \mathbb{R}^3$ and L maps into $\mathbb{R}^2$, the entire range is also $\mathbb{R}^2$: $L(\mathbb{R}^3) = \mathbb{R}^2$.

Example 13: Kernel of Differentiation

- Let D: $P_3 \to P_3$ be the differentiation operator, defined by $D(p(x)) = p'(x)$.
- In the textbook notation, P_3 consists of polynomials of degree at most 2. Write $p(x) = a + bx + cx^2$.

$$D(p(x)) = b + 2cx.$$

- The derivative is the zero polynomial if and only if $b = 0$ and $c = 0$. The constant a is arbitrary.

$$\ker(D) = \{p(x) = a : a \in \mathbb{R}\} = P_1.$$

- Thus, the kernel consists of all constant polynomials, including the zero polynomial.

Example 13: Range of Differentiation

- For $p(x) = a + bx + cx^2 \in P_3$, the derivative $b + 2cx$ belongs to P_2. Thus $D(P_3) \subseteq P_2$.
- Conversely, every polynomial $a + bx \in P_2$ has an antiderivative in P_3:

$$D\left(ax + \frac{b}{2}x^2\right) = a + bx.$$

- Therefore, every polynomial in P_2 occurs as an output of D, and

$$D(P_3) = P_2.$$

- Although the codomain is P_3, the range is its subspace P_2. This illustrates the distinction between the specified output space and the outputs that actually occur.

Basis Images and Matrix Representations

- Section 3.5 describes a vector by its coordinates in a basis. Section 4.1 describes a linear transformation by how it acts on vectors.
- Linearity means that the images of the basis vectors determine the image of every vector.

$$L(c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n) = c_1L(\mathbf{v}_1) + \cdots + c_nL(\mathbf{v}_n).$$

- Section 4.2 now uses coordinate vectors to represent this rule by matrix multiplication.

How Do Matrices Represent Linear Transformations?

Matrices Representing Linear Transformations

- Section 4.1 showed that each m × n matrix A defines a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ by multiplying each vector by A.
- We now establish the converse:

  every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ has a matrix representation.

$$L(\mathbf{x}) = A\mathbf{x}.$$

- For general finite-dimensional vector spaces, the matrix acts on coordinate vectors relative to specified ordered bases.

Theorem 4.2.1

If L is a linear transformation mapping $\mathbb{R}^n$ to $\mathbb{R}^m$, there is an $m \times n$ matrix A such that

$$L(\mathbf{x}) = A\mathbf{x}$$

for each $\mathbf{x} \in \mathbb{R}^n$. In fact, the jth column vector of A is given by

$$\mathbf{a}_j = L(\mathbf{e}_j), \ j = 1, 2, \ldots, n$$

- Each column records the image of one standard basis vector of the input space.

Proof of Theorem 4.2.1: Constructing A

For j = 1, …, n, define $a_j = L(e_j)$ and let

$$A = (a_{ij}) = (\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$$

Each $\mathbf{a}_j$ belongs to $\mathbb{R}^m$, so A has m rows and n columns. An arbitrary vector $\mathbf{x} \in \mathbb{R}^n$ has the standard basis expansion

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)^T = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \ldots + x_n\mathbf{e}_n$$

Linearity determines $L(\mathrm{x})$ from the images of these basis vectors.

Proof of Theorem 4.2.1: Verifying the Representation

$$
\begin{aligned}
L(\mathbf{x}) &= L(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \ldots + x_n\mathbf{e}_n) \\
&= x_1L(\mathbf{e}_1) + x_2L(\mathbf{e}_2) + \ldots + x_nL(\mathbf{e}_n) \\
&= x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n \\
&= (\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \\
&= A\mathbf{x}
\end{aligned}
$$

- The equality holds for every $\mathbf{x} \in \mathbb{R}^n$, so A represents L.
- Since the standard bases are used for both $\mathbb{R}^n$ and $\mathbb{R}^m$, A is called the standard matrix representation of L.

Example 1: Images of the Standard Basis Vectors

Let $L: \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $L(\mathbf{x}) = (x_1 + x_2, x_2 + x_3)^T$.

Find A such that $L(\mathbf{x}) = A\mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^3$.

$$\mathbf{a}_1 = L(\mathbf{e}_1) = L\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) = \begin{bmatrix}1+0\\0+0\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$$

$$\mathbf{a}_2 = L(\mathbf{e}_2) = L\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right) = \begin{bmatrix}0+1\\1+0\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$$

$$\mathbf{a}_3 = L(\mathbf{e}_3) = L\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right) = \begin{bmatrix}0+0\\0+1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$

$$A = (L(\mathbf{e}_1), L(\mathbf{e}_2), L(\mathbf{e}_3)) = \begin{bmatrix}1 & 1 & 0\\0 & 1 & 1\end{bmatrix}$$

$$A\mathbf{x} = \begin{bmatrix}1 & 1 & 0\\0 & 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}x_1+x_2\\x_2+x_3\end{bmatrix} = L(\mathbf{x}).$$

Thus, A is the standard matrix representation of L. Its three columns correspond to the three input coordinates. Its two rows correspond to the two output coordinates.

Example 2: Images under a Rotation

- Let L: $\mathbb{R}^2 \to \mathbb{R}^2$ which rotates each vector by an angle θ in the counterclockwise direction.
- Let $L(\mathbf{x}) = A\mathbf{x}$

  Since $\mathbf{a}_1 = L(\mathbf{e}_1) = (\cos\theta, \sin\theta)^T$, and $\mathbf{a}_2 = L(\mathbf{e}_2) = (-\sin\theta, \cos\theta)^T$

$$\Rightarrow \quad A = (\mathbf{a}_1, \mathbf{a}_2) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

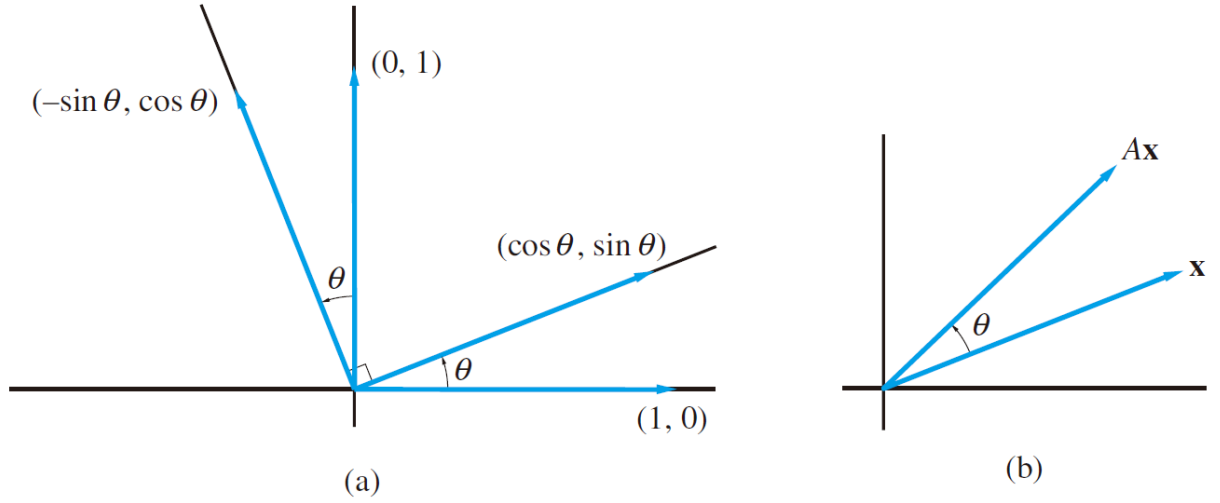


Figure 4.2.1.

Coordinates for a Transformation from V to W

- Let E = $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be an ordered basis for V and F = $\{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$ an ordered basis for W.
- Let L: V → W be linear. $\mathbf{v} = x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n, \qquad \mathbf{x} = [\mathbf{v}]_E.$
- We seek an m × n matrix A that sends input coordinates in E to output coordinates in F:

$$[L(\mathbf{v})]_F = A\mathbf{x} = A[\mathbf{v}]_E.$$

- The matrix acts on coordinates. The transformation L acts on the vectors in V.

Constructing the Columns for General Ordered Bases

- For each j = 1, …, n, express the image of v_j in the ordered basis F:

$$L(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + a_{2j}\mathbf{w}_2 + \ldots + a_{mj}\mathbf{w}_m$$

- The coefficients form column j of A:

$$\mathbf{a}_j = [L(\mathbf{v}_j)]_F = (a_{1j}, a_{2j}, \ldots, a_{mj})^T.$$

$$A = (a_{ij}) = (\mathbf{a}_1, \ldots, \mathbf{a}_n).$$

- The target basis has m vectors, so each coordinate column has m entries.

Deriving the Matrix Representation

- For $\mathbf{v} = x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n$, substitute the basis expansions of $L(\mathbf{v}_j)$:

$$L(\mathbf{v}) = L\left(\sum_{j=1}^{n} x_j \mathbf{v}_j\right) = \sum_{j=1}^{n}\left(x_j\ L(\mathbf{v}_j)\right) = \sum_{j=1}^{n}\left(x_j(\sum_{i=1}^{m} a_{ij}\mathbf{w}_i)\right)$$

$$= \sum_{i=1}^{m}\left(\left(\sum_{j=1}^{n} a_{ij}x_j\right)\mathbf{w}_i\right)$$

- Thus, with $y_i = \Sigma_j\, a_{ij}x_j$, the coordinate vector of $L(\mathbf{v})$ in F is $\mathbf{y} = A\mathbf{x}$. This proves the representation for every $\mathbf{v} \in V$.

Theorem 4.2.2: Matrix Representation Theorem

If $E = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $F = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\}$ are ordered bases for vector spaces V and W, respectively, then corresponding to each linear transformation L: $V \to W$ there is an $m \times n$ matrix A such that

$$[L(\mathbf{v})]_F = A\,[\mathbf{v}]_E \qquad \text{for each } \mathbf{v} \in V$$

A is the matrix representing L relative to the ordered bases E and F. In fact,

$$\mathbf{a}_j = [L(\mathbf{v}_j)]_{\boldsymbol{F}} \quad j = 1, 2, \ldots, n$$

Interpreting the Coordinate Representation

- x and y are the coordinate vectors of v and w relative to E and F.
- L maps v into w if and only if A maps x into y.

$$\mathbf{w} = L(\mathbf{v}) \iff [\mathbf{w}]_F = A[\mathbf{v}]_E.$$

$$\mathbf{x} = [\mathbf{v}]_E, \qquad \mathbf{y} = [\mathbf{w}]_F, \qquad \mathbf{y} = A\mathbf{x}.$$

- The chosen ordered bases determine how each vector is represented by its coordinate vector.

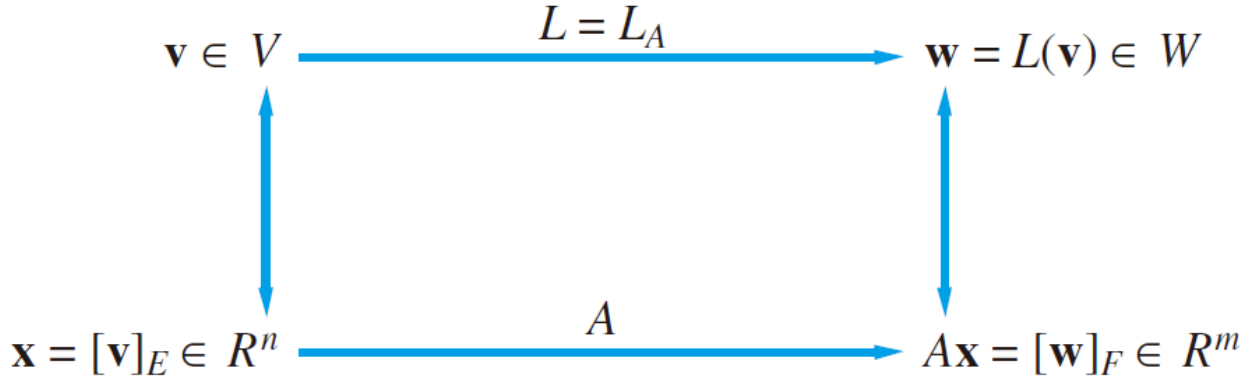


Figure 4.2.2.

Example 3: A Transformation Specified by Basis Vectors

Let L: $\mathbb{R}^3 \to \mathbb{R}^2$ be defined by

$$L(\mathbf{x}) = x_1\mathbf{b}_1 + (x_2 + x_3)\mathbf{b}_2, \qquad \mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Find the matrix A representing L with respect to the ordered bases $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $\{\mathbf{b}_1, \mathbf{b}_2\}$. The output coefficients must be read relative to $\{\mathbf{b}_1, \mathbf{b}_2\}$, in that order.

Example 3: Reading the Output Coordinates

$$L(\mathbf{e}_1) = L\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) = 1\cdot\mathbf{b}_1 + 0\cdot\mathbf{b}_2 \Rightarrow \mathbf{a}_1 = \left[L(\mathbf{e}_1)\right]_F = \begin{bmatrix}1\\0\end{bmatrix}$$

$$L(\mathbf{e}_2) = L\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right) = 0\cdot\mathbf{b}_1 + 1\cdot\mathbf{b}_2 \Rightarrow \mathbf{a}_2 = \left[L(\mathbf{e}_2)\right]_F = \begin{bmatrix}0\\1\end{bmatrix}$$

$$L(\mathbf{e}_3) = L\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right) = 0\cdot\mathbf{b}_1 + 1\cdot\mathbf{b}_2 \Rightarrow \mathbf{a}_3 = \left[L(\mathbf{e}_3)\right]_F = \begin{bmatrix}0\\1\end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 1\end{bmatrix}$$

Example 4: An Operator in a Nonstandard Basis

Use the ordered basis $\{\mathbf{b}_1, \mathbf{b}_2\}$ from Example 3. Let L: $\mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$L(a\,\mathbf{b}_1 + b\,\mathbf{b}_2) = (a+b)\mathbf{b}_1 + 2b\,\mathbf{b}_2$$

Find the matrix A representing L w. r. t. $\{\mathbf{b}_1, \mathbf{b}_2\}$.

- $L(\mathbf{b}_1) = L(1\mathbf{b}_1 + 0\mathbf{b}_2) = (1+0)\mathbf{b}_1 + (2\times 0)\mathbf{b}_2$

  $= 1\mathbf{b}_1 + 0\mathbf{b}_2 \Rightarrow \mathbf{a}_1 = [L(\mathbf{b}_1)]_{\boldsymbol{F}} = (1, 0)^T$

- $L(\mathbf{b}_2) = L(0\mathbf{b}_1 + 1\mathbf{b}_2) = (0+1)\mathbf{b}_1 + (2\times 1)\mathbf{b}_2$

  $= 1\mathbf{b}_1 + 2\mathbf{b}_2 \Rightarrow \mathbf{a}_2 = [L(\mathbf{b}_2)]_{\boldsymbol{F}} = (1, 2)^T$

$$\Rightarrow A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$$

Example 5: Matrix Representation of Differentiation

Let D: $P_3 \to P_2$ be defined by $D(p) = p'$. Use the ordered bases $[x^2, x, 1]$ for P_3 and $[x, 1]$ for P_2.

$$D(x^2) = 2x + 0 \cdot 1$$
$$D(x) = 0x + 1 \cdot 1$$
$$D(1) = 0x + 0 \cdot 1$$

The coordinate vectors $(2, 0)^T$, $(0, 1)^T$, and $(0, 0)^T$ form the columns:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Example 5: Applying the Matrix to a Polynomial

- If $p(x) = ax^2 + bx + c$, its coordinate vector with respect to $[x^2, x, 1]$ is $(a, b, c)^T$.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2a \\ b \end{bmatrix}$$

- The output coordinates are relative to $[x, 1]$. Therefore,

$$D(ax^2 + bx + c) = 2ax + b$$

- The 2×3 matrix maps three input coefficients to two output coefficients, matching $\dim P_3 = 3$ and $\dim P_2 = 2$.

Theorem 4.2.3

Let $E = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ and $F = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m\}$ be ordered bases for $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. If $L: \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation and A is the matrix representing L with respect to E and F, then

$$\mathbf{a}_j = B^{-1}L(\mathbf{u}_j) \quad \text{for } j = 1, 2, \ldots, n$$

where $B = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m)$.

Proof of Theorem 4.2.3

- The jth column of A is the coordinate vector of $L(\mathbf{u}_j)$ in the target basis F. Thus,

$$\begin{aligned} L(\mathbf{u}_j) &= a_{1j}\mathbf{b}_1 + a_{2j}\mathbf{b}_2 + \cdots + a_{mj}\mathbf{b}_m \\ &= B\mathbf{a}_j. \end{aligned}$$

- The matrix B is nonsingular because its columns form a basis for $\mathbb{R}^m$.

  Multiplying by B^{-1} gives

$$\mathbf{a}_j = B^{-1}L(\mathbf{u}_j), \qquad j = 1, \ldots, n.$$

- Thus, finding each output coordinate vector is equivalent to solving $B\,\mathbf{a}_j = L(\mathbf{u}_j)$.

Corollary 4.2.4

If A is the matrix representing the linear transformation L: $R^n \rightarrow R^m$ with respect to the bases

$$E = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n] \text{ and } F = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m],$$

then the reduced row echelon form of $(\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m \mid L(\mathbf{u}_1), \ldots, \boldsymbol{L}(\mathbf{u}_n))$ is $(I \mid A)$.

- One row reduction therefore finds all n columns of the matrix representation at once.

Proof of Corollary 4.2.4

Let B = ($\mathbf{b}_1, \ldots, \mathbf{b}_m$). Since B is nonsingular, the augmented matrix is row equivalent to B^{-1} ($B|\ L(\mathbf{u}_1), \ldots, L(\mathbf{u}_n)$)

$$= (\ I \mid B^{-1}L(\mathbf{u}_1), \ldots, \mathrm{B}^{-1}L(\mathbf{u}_n))$$

$$= (\ I \mid \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$$

$$= (\ I \mid A)$$

Theorem 4.2.3 identifies the right-hand columns as the columns of A. The identity block supplies a leading 1 in every row, so (I | A) is in reduced row echelon form.

Example 6: Domain and Codomain Bases

Let L: $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $L(\mathbf{x}) = (x_2, x_1 + x_2, x_1 - x_2)^T$. Find its matrix representation for the ordered bases $\{\mathbf{u}_1, \mathbf{u}_2\}$ and $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$, where

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \ \mathbf{u}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \text{ and } \mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \ \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \ \mathbf{b}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

- First evaluate L at each domain basis vector:

$$L(\mathbf{u}_1) = L\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1+2 \\ 1-2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad L(\mathbf{u}_2) = L\left(\begin{bmatrix} 3 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 3+1 \\ 3-1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$$

Example 6: Row Reduction for the Coordinate Columns and Verification

$$(B \mid L(\mathbf{u}_1)\, L(\mathbf{u}_2)) = \left[\begin{array}{ccc|cc} 1 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3 & 4 \\ 0 & 0 & 1 & -1 & 2 \end{array}\right] \rightarrow \left[\begin{array}{ccc|cc} 1 & 0 & 0 & -1 & -3 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 1 & -1 & 2 \end{array}\right]$$

$$\Rightarrow A = \begin{bmatrix} -1 & -3 \\ 4 & 2 \\ -1 & 2 \end{bmatrix}$$

- Verify the first and second column by reconstructing $L(\mathbf{u}_1)$ and $L(\mathbf{u}_2)$:

$$L(\mathbf{u}_1) = -\mathbf{b}_1 + 4\mathbf{b}_2 - \mathbf{b}_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad L(\mathbf{u}_2) = -3\mathbf{b}_1 + 2\mathbf{b}_2 + 2\mathbf{b}_3 = \begin{bmatrix} -3 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$$

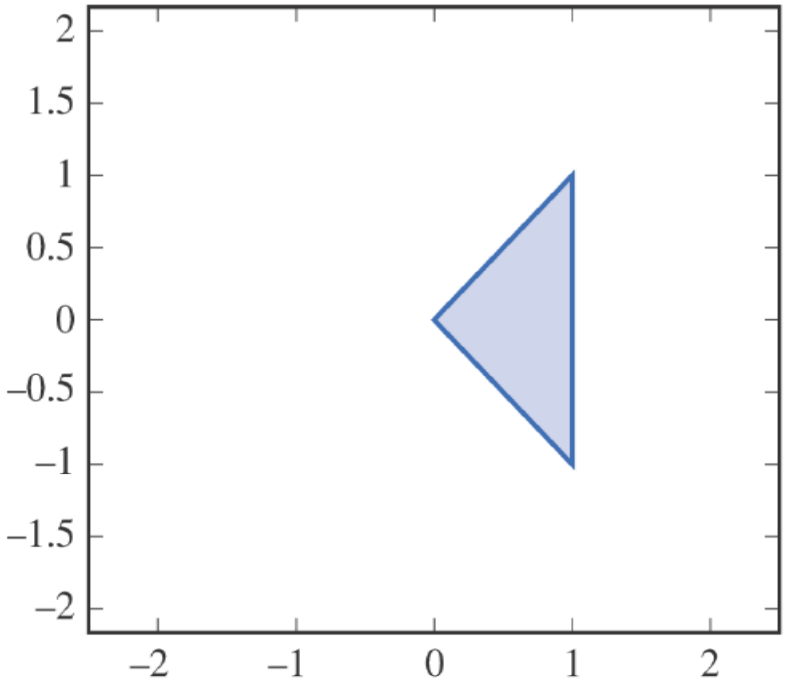
- Each column gives the correct output coordinates in the specified codomain basis.

Application 1: Storing a Picture as a Matrix

- A picture in the plane can be stored as a sequence of vertices. Their x-coordinates form the first row and their y-coordinates the second. Connect successive points by straight lines.
- For the triangle with vertices (0, 0), (1, 1), and (1, −1), use

$$T = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$

- The last column repeats the first vertex to close the triangle.



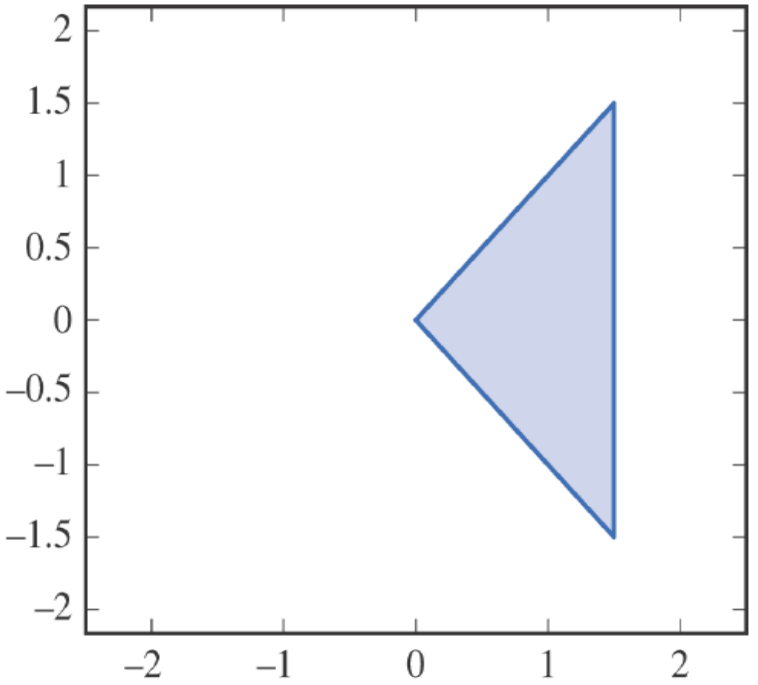
(a) Triangle defined by T

Application 1: Dilations and Contractions

- A linear operator L(x) = cx is a dilation if c > 1 and a contraction if 0 < c < 1. Its matrix is cI, where I is the 2 × 2 identity matrix.

$$1.5T = \begin{bmatrix} 0 & 1.5 & 1.5 & 0 \\ 0 & 1.5 & -1.5 & 0 \end{bmatrix}.$$

- Multiplying every column of T by 1.5 enlarges the triangle by that factor.



(b) Dilation by factor of 1.5

Application 1: Reflection about the x-Axis

- Let L_x reflect a vector about the x-axis. Its action on the standard basis is

$$L_x(\mathbf{e}_1) = \mathbf{e}_1, \qquad L_x(\mathbf{e}_2) = -\mathbf{e}_2.$$

- Therefore, the matrix representation is

$$A = (L_x(\mathbf{e}_1), L_x(\mathbf{e}_2)) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}.$$

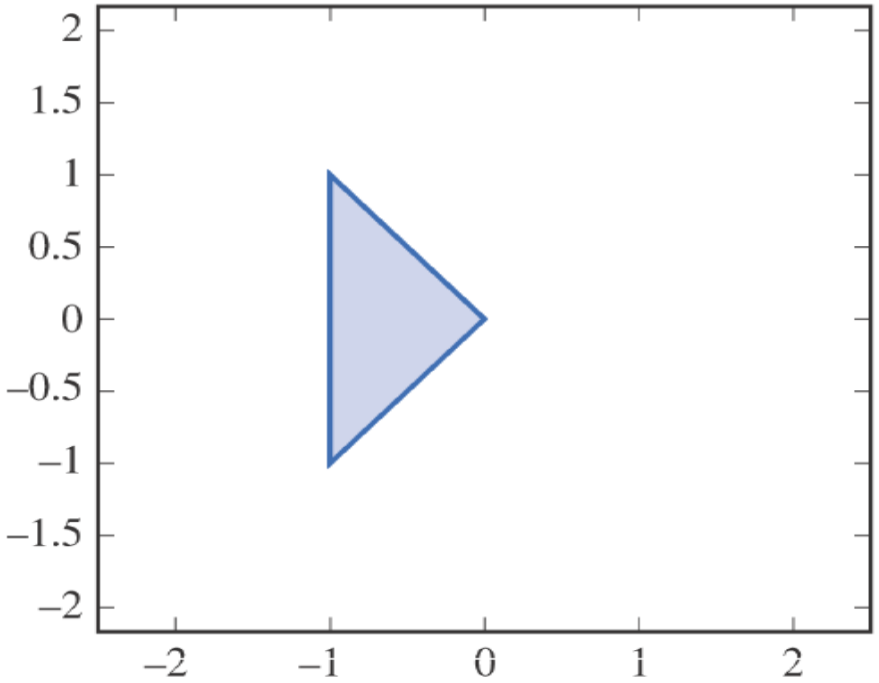
- The first coordinate stays the same and the second changes sign.

Application 1: Reflection about the y-Axis

- For reflection about the y-axis, $L_\gamma(e_1) = -e_1$ and $L_\gamma(e_2) = e_2$.
- Hence

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad AT = \begin{bmatrix} 0 & -1 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

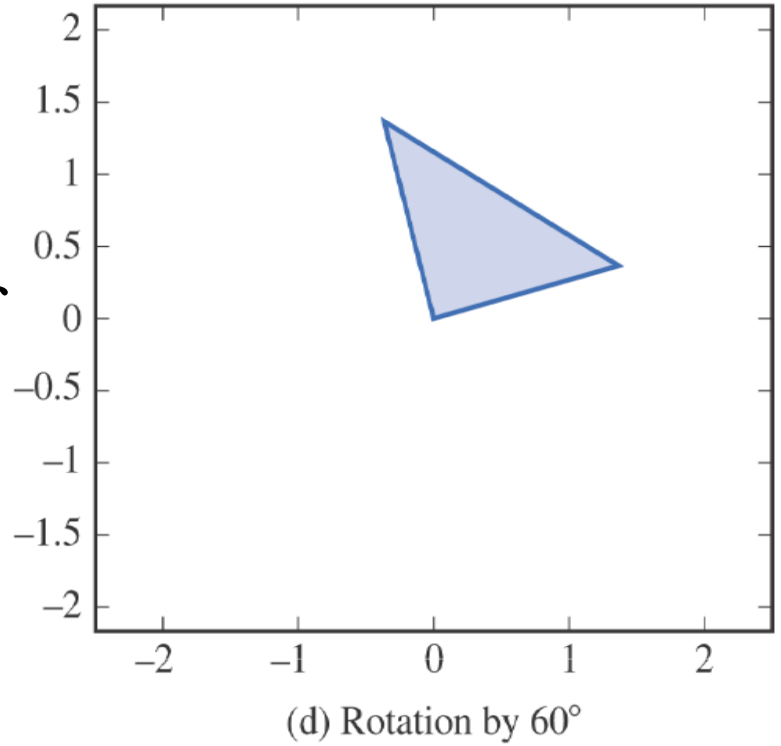
- The columns of AT are the reflected vertices.



(c) Reflection about *y*-axis

Application 1: Rotation of the Triangle

- A counterclockwise rotation about the origin through θ has matrix $$A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$$



(d) Rotation by 60°

- For θ = 60°, multiply T by this matrix. The columns of AT are the rotated vertices.

$$AT=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}0 & 1 & 1 & 0\\ 0 & 1 & -1 & 0\end{bmatrix}$$

$$=\begin{bmatrix}0 & \cos\theta-\sin\theta & \cos\theta+\sin\theta & 0\\ 0 & \sin\theta+\cos\theta & \sin\theta-\cos\theta & 0\end{bmatrix}$$

Application 1: Translations

- A translation by a vector a has the form

$$L(\mathbf{x}) = \mathbf{x} + \mathbf{a}.$$

- If $\mathbf{a} \neq \mathbf{0}$, this mapping is not a linear transformation. In particular,

$$L(\mathbf{0}) = \mathbf{a} \neq \mathbf{0}.$$

- Consequently, no 2×2 matrix A can satisfy $A\mathbf{x} = \mathbf{x} + \mathbf{a}$ for every $\mathbf{x} \in \mathbb{R}^2$.
- Computer graphics uses homogeneous coordinates to carry out translations by matrix multiplication.

Application 1: Homogeneous Coordinates

- In homogeneous coordinates, represent $(x_1, x_2)^T$ by the vector $(x_1, x_2, 1)^T$. When plotting the point, ignore the third coordinate.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

- Replace a 2×2 matrix by a 3×3 matrix that retains its first two rows and columns and attaches the third row and column of the 3×3 identity matrix.

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3x_1 \\ 3x_2 \\ 1 \end{bmatrix}$$

Application 1: A Translation Matrix

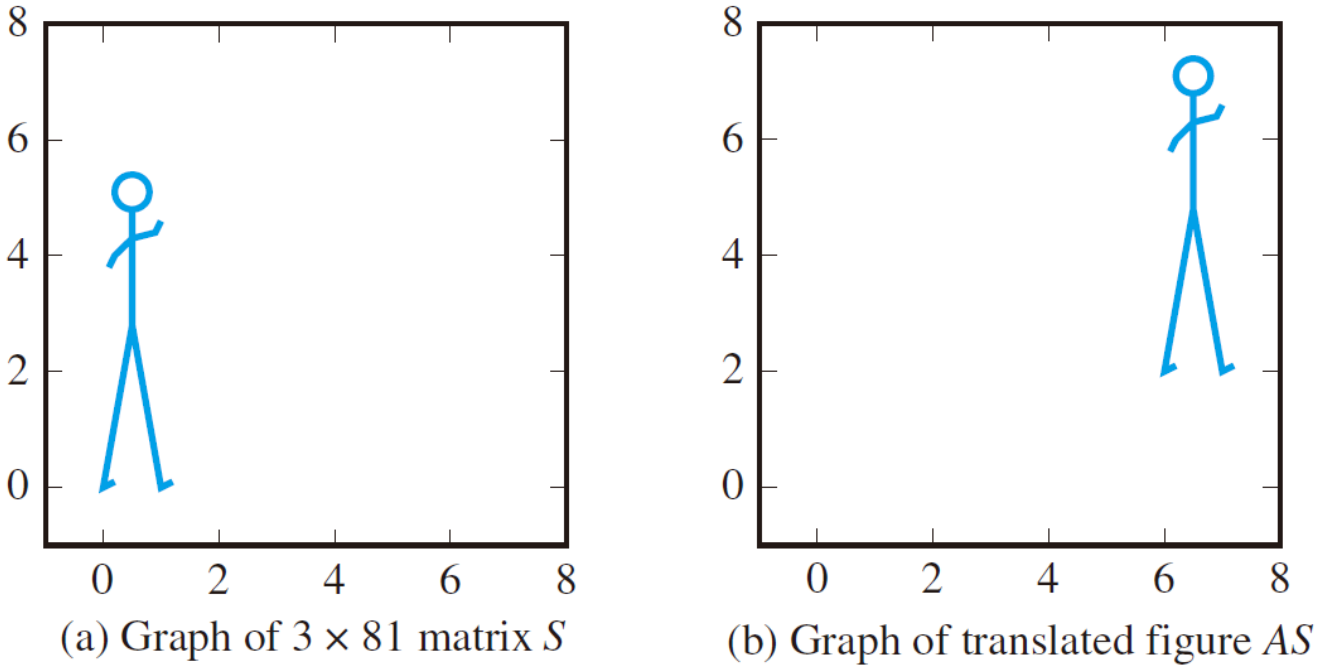
- For a translation by $a = (6, 2)^T$, replace the first two entries of the third column of the 3×3 identity matrix by 6 and 2:

$$\begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 + 6 \\ x_2 + 2 \\ 1 \end{bmatrix}$$

- This 3×3 matrix defines a linear transformation on $\mathbb{R}^3$. On vectors with third coordinate 1, it produces the required translation in the plane.

Application 1: Translating a Stored Figure

- Figure 4.2.4(a) shows a stick figure generated from a 3 × 81 matrix S. Each column contains homogeneous coordinates of one stored point. With the preceding translation matrix A, the columns of AS give the translated figure in Figure 4.2.4(b).
- The same matrix multiplication transforms every stored point. Successive drawings create the effect of animation.



(a) Graph of 3 × 81 matrix S (b) Graph of translated figure AS

Figure 4.2.4.

How Does a Change of Basis Relate Two Matrix Representations?

Different Matrices for the Same Linear Operator

- If L is a linear operator on an n-dimensional vector space V, the matrix representation of L depends on the ordered basis chosen for V.
- Using different bases can produce different $n \times n$ matrices representing the same operator. This section characterizes the relationship between these matrices.
- Section 3.5 gives the transition matrix for changing coordinates. Section 4.2 gives the matrix of an operator in a chosen basis. Here we combine these two ideas.

Introductory Example: The Standard Matrix

- Let L be the linear transformation mapping $\mathbb{R}^2$ into itself defined by

$$L(\mathbf{x}) = (2x_1, x_1 + x_2)^T.$$

- Apply L to the standard basis vectors. Their images become the columns of the representing matrix:

$$L(\mathbf{e}_1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad L(\mathbf{e}_2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}.$$

- Thus, $L(\mathbf{x}) = A\mathbf{x}$, and A represents L with respect to $\{\mathbf{e}_1, \mathbf{e}_2\}$.

Introductory Example: Images of a Different Basis

- Use the new basis vectors $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
- Their images in standard coordinates are

$$L(\mathbf{u}_1) = A\mathbf{u}_1 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \qquad L(\mathbf{u}_2) = A\mathbf{u}_2 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

- To construct the new representing matrix, express these images as linear combinations of $\mathbf{u}_1$ and $\mathbf{u}_2$.

Introductory Example: The Transition Matrices

- The transition matrix from $\{\mathbf{u}_1, \mathbf{u}_2\}$ to $\{\mathbf{e}_1, \mathbf{e}_2\}$ has the new basis vectors as columns:

$$U = (\mathbf{u}_1, \mathbf{u}_2) = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

- Its determinant is 2, so the vectors form a basis and U is nonsingular.

  The reverse transition matrix is

$$U^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

- Multiplication by U^{-1} converts standard coordinates into coordinates with respect to $\{\mathbf{u}_1, \mathbf{u}_2\}$.

Introductory Example: The New Representing Matrix

$$U^{-1}L(\mathbf{u}_1) = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \qquad U^{-1}L(\mathbf{u}_2) = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- Therefore, $L(\mathbf{u}_1) = 2\mathbf{u}_1 + 0\mathbf{u}_2$ and $L(\mathbf{u}_2) = -\mathbf{u}_1 + \mathbf{u}_2$. These coordinate vectors form the columns of

$$B = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$$

- The operator is the same, but its representing matrix changes with the basis.

Introductory Example: The Relation $B = U^{-1}AU$

- The two columns of B are $U^{-1}A\mathbf{u}_1$ and $U^{-1}A\mathbf{u}_2$. Hence

$$B = (U^{-1}A\mathbf{u}_1, U^{-1}A\mathbf{u}_2) = U^{-1}A(\mathbf{u}_1, \mathbf{u}_2) = U^{-1}AU.$$

$$U^{-1}AU = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}.$$

- A represents L in the standard basis. B represents L in $\{\mathbf{u}_1, \mathbf{u}_2\}$.

  U changes coordinates from $\{\mathbf{u}_1, \mathbf{u}_2\}$ to the standard basis.

Theorem 4.3.1: A Change of Representing Matrix

Let $E = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $F = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\}$ be two ordered bases for a vector space V and let L be a linear operator on V. Let S be the transition matrix representing the change from F to E. If A is the matrix representing L w. r. t. E and B is the matrix representing L w. r. t. F, then $B = S^{-1}AS$.

- The direction of S is from F to E. Its inverse changes coordinates from E to F.

Proof of Theorem 4.3.1: Coordinates in Two Bases

- Let $\mathbf{x}$ be any vector in R^n and let $\mathbf{x} = [\mathbf{v}]_F$

$$\mathbf{v} = x_1\mathbf{w}_1 + x_2\mathbf{w}_2 + \ldots + x_n\mathbf{w}_n$$

- Since S is the transition matrix representing the change from F to E

  Let

$$\mathbf{y} = S\mathbf{x},\ \mathbf{t} = A\mathbf{y},\ \mathbf{z} = B\mathbf{x}$$

- Let $\mathbf{x}$ be any vector in R^n and let $\mathbf{y} = [\mathbf{v}]_E$

$$\mathbf{v} = y_1\mathbf{v}_1 + y_2\mathbf{v}_2 + \ldots + y_n\mathbf{v}_n$$

Proof of Theorem 4.3.1: Transforming the Coordinates

- Since A represents L w. r. t. E and B represents L w. r. t. F, we have

$$\mathbf{t} = [L(\mathbf{v})]_E \text{ and } \mathbf{z} = [L(\mathbf{v})]_F$$

- The transition matrix from E to F is S^{-1}. Consequently,

$$S^{-1}\mathbf{t} = \mathbf{z}$$

- Substituting $\mathbf{y} = S\mathbf{x}$, $\mathbf{t} = A\mathbf{y}$, and $\mathbf{z} = B\mathbf{x}$ gives

$$S^{-1}AS\mathbf{x} = S^{-1}A\mathbf{y} = S^{-1}\mathbf{t} = \mathbf{z} = B\mathbf{x}.$$

Proof of Theorem 4.3.1: Equality of the Matrices

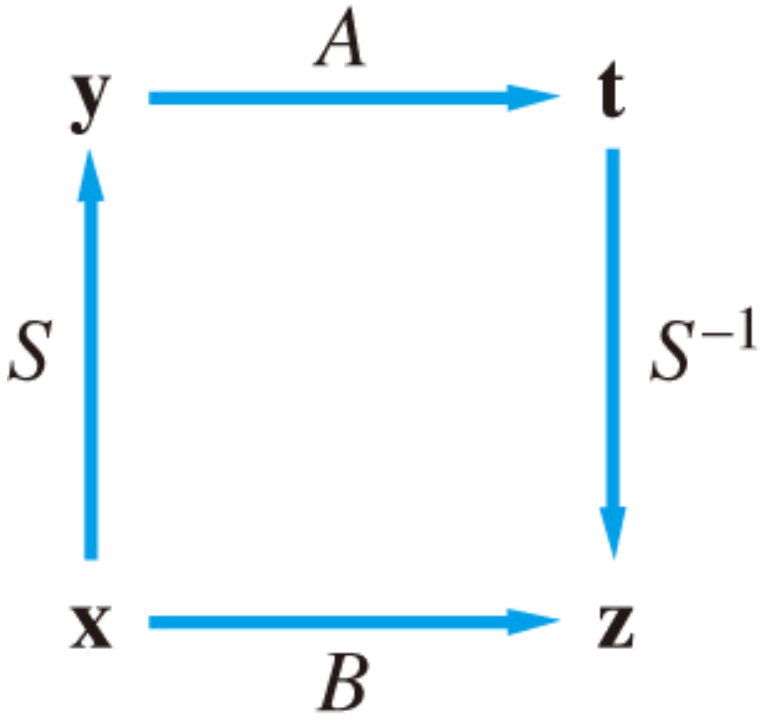
- The preceding identity holds for every $\mathbf{x} \in \mathbb{R}^n$:

$$S^{-1}AS\mathbf{x} = B\mathbf{x}$$

- In particular, set x equal to each standard basis vector e_j.
- Then the jth columns of $S^{-1}AS$ and B are equal, for every $j = 1, \ldots, n$.

$$S^{-1}AS\mathbf{x} = B$$

- Thus, changing the basis changes the representing matrix by this product. The linear operator L itself remains the same.



The Identity Transformation and Change of Basis

- The transition matrix S represents the identity transformation I relative to the input basis F and the output basis E. Its inverse represents I relative to E and F.
- The same operator L can be written as the composition I◦ L ◦ I. Starting with coordinates in F, the corresponding matrix product is $B = S^{-1}AS$.

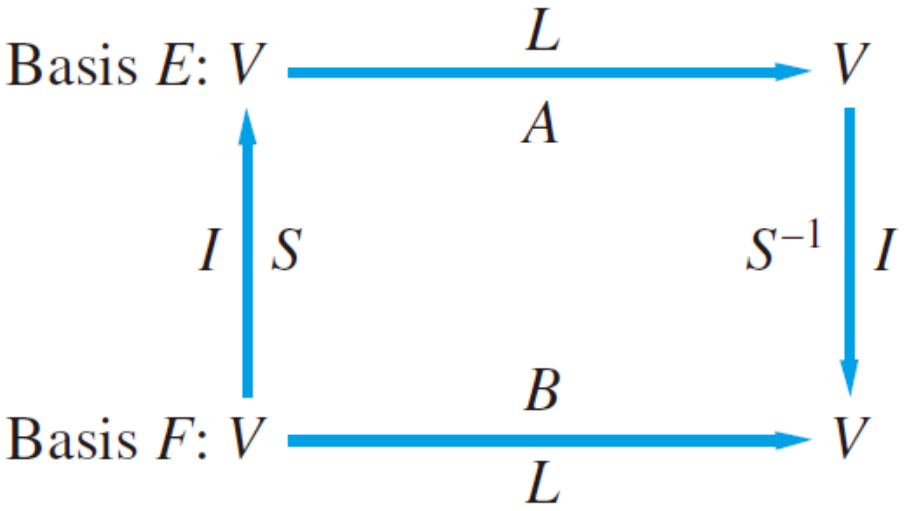


Figure 4.3.2.

The rightmost factor S acts first, A applies L in basis E, and S^{-1} returns the output coordinates to basis F.

Similar Matrices

Let A and B be two $n \times n$ matrices. B is said to be **similar** to A if there exists a nonsingular matrix S such that $B = S^{-1}AS$.

- Similarity is symmetric. Multiplying on the left by S and on the right by S^{-1} gives $A = SBS^{-1} = (S^{-1})^{-1}B(S^{-1})$
- Thus, A is similar to B, and we may simply say that A and B are similar matrices.

Similarity and Representations of One Operator

- Theorem 4.3.1 shows that matrices representing the same linear operator in two ordered bases are similar.
- Conversely, suppose A represents L with respect to $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $B = S^{-1}AS$ for a nonsingular matrix S. Define

$$\mathbf{w}_j = s_{1j}\mathbf{v}_1 + s_{2j}\mathbf{v}_2 + \cdots + s_{nj}\mathbf{v}_n, \qquad j = 1, \ldots, n.$$

- As in §3.5, nonsingularity of S implies that $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ is a basis. The matrix S changes coordinates from this basis to $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$.
- Theorem 4.3.1 then shows that B represents L in the ordered basis $\{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$.

The Converse: Why the New Vectors Form a Basis

- Suppose a linear combination of the vectors defined by the columns of S is zero.
- Substituting their definitions gives $$\sum_{j=1}^{n} x_j \mathbf{w}_j = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} s_{ij} x_j \right) \mathbf{v}_i = \mathbf{0}.$$
- Since $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent, every coefficient in parentheses is zero.
- Equivalently, $S\mathbf{x} = 0$.
- Nonsingularity of S gives $\mathbf{x} = \mathbf{0}$, so $w_1, \ldots, w_n$ are linearly independent. There are n of them in the n-dimensional space V. Therefore, they form a basis, and S is the transition matrix from this basis to $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$.

Example 1: Differentiation in Two Polynomial Bases

Let D be the differentiation operator on P_3. Find the matrix B representing D with respect to $[1, x, x^2]$ and the matrix A representing D with respect to $[1, 2x, 4x^2 - 2]$.

- First use the standard polynomial basis [1, x, x²]:

$$D(1) = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2$$
$$D(x) = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2$$
$$D(x^2) = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2$$

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

- Each column contains the coordinates of the derivative of the corresponding basis element.

Example 1: The Second Representing Matrix

- Apply D to the ordered basis $[1, 2x, 4x^2 - 2]$ and express the derivatives in the same basis:

$$D(1) = 0 \cdot 1 + 0 \cdot 2x + 0 \cdot (4x^2 - 2)$$
$$D(2x) = 2 \cdot 1 + 0 \cdot 2x + 0 \cdot (4x^2 - 2)$$
$$D(4x^2 - 2) = 0 \cdot 1 + 4 \cdot 2x + 0 \cdot (4x^2 - 2)$$

$$A = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

- The operator D is unchanged. The coordinate vectors of its basis images depend on the ordered basis.

Example 1: The Transition Matrix and Its Direction

- As in Example 6 of §3.5, the transition matrix from $[1, 2x, 4x^2 - 2]$ to $[1, x, x^2]$ is

$$S = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \quad \text{and} \quad S^{-1} = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}$$

- Here B represents D in $[1, x, x^2]$, and A represents D in $[1, 2x, 4x^2 - 2]$. Therefore the similarity relation with the textbook's labels is

$$A = SBS^{-1}$$

- S changes the input coordinates to the standard polynomial basis, B applies differentiation, and S^{-1} returns the result to the other basis.

Example 1: Verification by Matrix Multiplication

$$BS = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$S^{-1}BS = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} = A.$$

- This agrees with the matrix constructed directly from $D(1)$, $D(2x)$, and $D(4x^2 - 2)$.

Example 2: A Simpler Representing Matrix

$L: \mathbb{R}^3 \to \mathbb{R}^3$ defined by $L(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}$$

Thus, the matrix A represents L with respect to $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, find the matrix representing L with respect to $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ where

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \mathbf{y}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Example 2: Images of the Three Basis Vectors

$$L(\mathbf{y}_1) = A\mathbf{y}_1 = \begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0\mathbf{y}_1 + 0\mathbf{y}_2 + 0\mathbf{y}_3$$

$$L(\mathbf{y}_2) = A\mathbf{y}_2 = \begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = 0\mathbf{y}_1 + 1\mathbf{y}_2 + 0\mathbf{y}_3$$

$$L(\mathbf{y}_3) = A\mathbf{y}_3 = \begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = 0\mathbf{y}_1 + 0\mathbf{y}_2 + 4\mathbf{y}_3$$

- Each image is already a scalar multiple of its own basis vector, so the coordinates are immediate.

Example 2: A Diagonal Representation

- The coordinate vectors of these images give the columns of

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

- Thus, D represents the same operator L with respect to $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$. It is simpler than the standard matrix A.

Example 2: The Transition Matrix Y

- The transition matrix from $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$ to the standard basis is

$$S = (\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3) = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

- Its determinant is $-3 \neq 0$, confirming that the three vectors form a basis. Direct row reduction gives

$$S^{-1} = \begin{bmatrix} 0 & -1 & 1 \\ \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

- Theorem 4.3.1 gives $D = S^{-1}AS$. The next calculation verifies the matrix obtained from the basis images.

Example 2: Verification of $D = S^{-1}AS$

$$S^{-1}AS = \begin{bmatrix} 0 & -1 & 1 \\ \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} = D$$

- The direct basis-image method and the transition-matrix method give the same diagonal matrix.

Why a Simpler Representation Is Useful

- In Example 2, L is represented by a diagonal matrix $\mathbf{D}$ in the basis $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$.
- Multiplication by D acts separately on the three coordinates.

$$D\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \\ 4x_3 \end{bmatrix}, \qquad D^n\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \\ 4^n x_3 \end{bmatrix} \quad (n \geq 1).$$

- It is easier to compute $D\mathbf{x}$ and $D^n\mathbf{x}$ than to repeatedly multiply by the original matrix A. A suitable basis makes the action of the operator easier to understand.

Thank You!